

I N D I G E N E

GROSS ANATOMY OF THE UPPER LIMB

ANA 217 — Complete Exam-Ready Compendium

Osteology · Joints & Ligaments · Musculature · Vasculature · Axilla & Lymphatics · Master Hack Sheet

200 Level — Nursing Science, Federal University Oye-Ekiti (FUOYE)
INDIGENE Study Console · Updated Edition — full upper-limb mastery

TABLE OF CONTENTS

1. Anatomical Planes, Axes & Terminology

Anatomical Position · Planes & Axes (with diagrams) · Terms of Relationship · Terms of Movement

2. Osteology of the Upper Limb

Clavicle · Scapula · Humerus · Radius & Ulna · Bones of the Wrist and Hand

3. Joints and Ligaments of the Upper Limb

Sternoclavicular Joint · Other Clavicular Ligaments · Glenohumeral Joint (incl. Bursae) · Elbow Joint · Wrist Joint · Carpometacarpal & Metacarpophalangeal Joints · Interphalangeal Joints

4. Nerve Supply — The Brachial Plexus

Roots · Trunks · Divisions · Cords · Terminal Branches (with diagrams) · Erb's & Klumpke's Palsy

5. Musculature of the Upper Limb

Scapular Stabilizers · Rotator Cuff & Shoulder Movers · Thoracic Wall Muscles · Scapular Movements · Arm (Ant/Post) · Forearm (Ant/Post) · Intrinsic Hand Muscles

6. Vasculature of the Upper Limb

Subclavian · Axillary · Brachial · Ulnar · Radial Arteries · Arteries of the Hand · Venous Drainage

7. Lymphatic Drainage of the Upper Limb

Cubital Nodes · Axillary Node Groups · Shared Drainage With the Breast

8. The Axilla

Apex, Base & Four Walls (with diagram) · Contents · the NAVEL Rule

9. Surface Anatomy of the Upper Limb

Palpable Bony Points · Pulses · Veins for Cannulation

10. Tutorial Preview — Cubital Fossa, Carpal Tunnel & Snuffbox

Boundaries, Contents and the Exam Traps for Each Space

11. Master Hack Sheet

Every mnemonic, trap, and high-yield rule compressed onto one section — for last-minute review

1. ANATOMICAL PLANES, AXES & TERMINOLOGY

Foundational vocabulary — every other section in this compendium assumes you know these terms cold. If you can't answer a T/F question on planes or terms, nothing else in anatomy will make sense either.

The Anatomical Position

Alzheimer's Mode: Before you can describe ANY direction in the body, you need a fixed reference pose — like "north" only makes sense once you have a map orientation. The anatomical position is that fixed map.

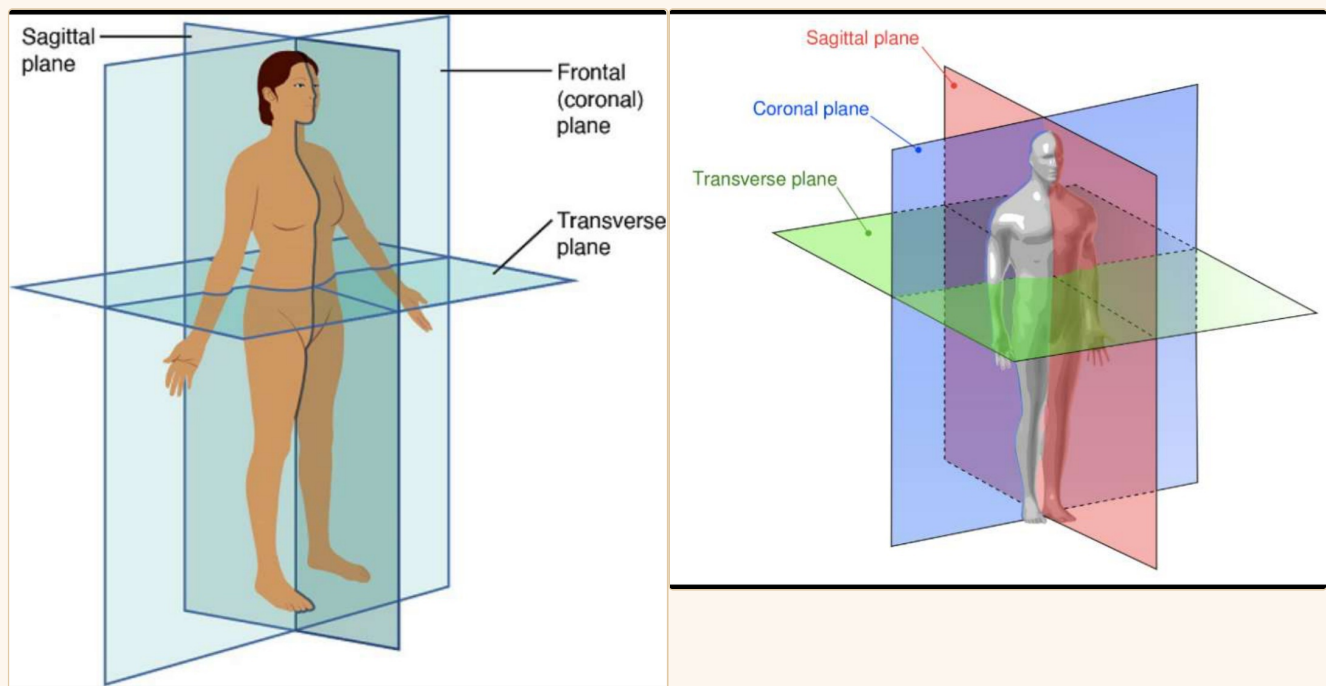
- Standing erect, facing forward, head/eyes/toes pointing forward
- Arms at the sides, **palms facing forward (supinated)** — the detail everyone forgets
- Legs together, feet flat, toes forward

EXAM HACK

Every anatomical description — planes, directional terms, movements — is stated **as if the body is always in this position**, even if the real body is lying down or moving. Because the anatomical position starts with palms forward, this IS the fully supinated reference point — any backward turn from here is pronation.

The Three Planes and Their Axes

Alzheimer's Mode: Picture slicing a loaf of bread three different directions. Each cut only allows movement to swing around the axis running perpendicular to it — like a door swinging on its hinge.



The three anatomical planes intersecting through the body in anatomical position.

Plane	Divides body into	Axis it rotates around
Sagittal (vertical, front-to-back)	Left + Right	Transverse axis (side-to-side)
Coronal/Frontal (vertical, side-to-side)	Anterior + Posterior	Sagittal (AP) axis (front-to-back)
Transverse/Axial (horizontal)	Superior + Inferior	Longitudinal axis (vertical)

- **Median (midsagittal) plane** — the ONE sagittal plane through the exact midline → equal left/right halves.
- **Parasagittal plane** — any other sagittal plane, off-center → unequal left/right portions.

EXAM HACK — "PERPENDICULAR PARTNERS"

The axis a movement rotates around is always perpendicular to the plane the movement happens in. Flexion/extension (sagittal plane — elbow, shoulder bending) rotate around a **transverse axis**. Abduction/adduction (coronal plane) rotate around a **sagittal axis**. Rotation/pronation-supination (transverse plane) rotate around a **longitudinal axis**. This single table answers most "which axis" questions instantly.

Only the **median plane** gives exactly equal halves — any other sagittal cut (parasagittal) does NOT. A classic trap states "the sagittal plane always divides the body into two equal halves" — false, unless it specifies the median plane.

Terms of Relationship (Position/Location)

Term pair	Meaning
Superior / Inferior	Toward the head / toward the feet
Anterior / Posterior (ventral/dorsal)	Toward the front / toward the back
Medial / Lateral	Toward the median plane / away from it
Proximal / Distal	Closer to / farther from the trunk (limbs only)
Superficial / Deep	Closer to / farther from the skin surface
Ipsilateral / Contralateral	Same side / opposite side of the body

EXAM HACK

Relationship terms are always **relative comparisons between two specific structures** — never an absolute label on one structure alone. A structure can be lateral to one thing and medial to another at the same time (e.g. the ulna is medial to the radius but lateral to structures on the wrist's ulnar side). Also: **proximal/distal has nothing to do with the midline** — that's medial/lateral's job. A common trap conflates "proximal" with "medial," which is wrong; proximal is strictly about distance from the trunk along a limb.

Terms of Movement

Term	Meaning
Flexion / Extension	Decreases / increases the angle between two bones (sagittal plane)
Abduction / Adduction	Away from / toward the median plane (coronal plane) — for fingers/toes, reference is the middle finger, not the body midline
Medial / Lateral rotation	Anterior surface of a limb turns toward / away from the median plane (transverse plane)
Circumduction	A COMBINATION — sequential flexion→abduction→extension→adduction, tracing a cone; NOT a separate movement type of its own
Pronation / Supination	Forearm-specific — palm turns posteriorly / anteriorly
Elevation / Depression	Structure moves superiorly/inferiorly — occurs at the scapula AND the mandible, NOT exclusively the glenohumeral joint
Protraction / Retraction	Structure moves anteriorly/posteriorly (scapula)
Opposition	Bringing the thumb into contact with the fingertips (hand only)

EXAM HACK — COMMON T/F TRAPS

- Circumduction "does not involve flexion/extension/abduction/adduction" — FALSE, it's built entirely from those four in sequence.
- Elevation/depression "only at the shoulder joint" — FALSE, also occurs at the mandible (TMJ), and shoulder elevation is really a **scapular** movement, not glenohumeral.
- Abducting the arm to 90° "necessarily involves rotation" — FALSE, pure 90° abduction is a coronal-plane movement alone; rotation only becomes necessary for abduction beyond 90° (scapulohumeral rhythm).
- In anatomical position, "palms face posteriorly" — FALSE, they face **anteriorly** (this is the fully supinated position).

CLINICAL — WHY THIS MATTERS

Every fracture description, muscle action, and movement test in this entire compendium is written using this vocabulary. "Medial epicondyle," "posterior circumflex humeral artery," "abduction 15–90°" — none of it means anything without a rock-solid grip on planes, axes, and these directional/movement terms. Master this section first; everything after it becomes dramatically easier to hold onto.

2. OSTEOLOGY OF THE UPPER LIMB

The bony skeleton of the upper limb comprises the **pectoral girdle** (scapula + clavicle), the **arm** (humerus), the **forearm** (radius + ulna), and the **hand** (carpals, metacarpals, phalanges). One upper limb has **32 bones total**: clavicle(1) + scapula(1) + humerus(1) + radius(1) + ulna(1) + carpals(8) + metacarpals(5) + phalanges(14). Unlike the lower limb — rigidly bolted to the axial skeleton through the pelvis for weight-bearing — the upper limb is only loosely slung from the trunk by muscles, trading stability for the huge mobility and fine manipulation the hand needs.

Clavicle

Alzheimer's Mode: The clavicle is the only long bone lying horizontally, shaped like a stretched-out italic "f" or elongated S — convex forward medially, concave forward laterally. It acts as a strut holding the shoulder out and away from the trunk.

CORRECTION — OSSIFICATION

The clavicle is the **only bone in the body that ossifies by BOTH mechanisms**: the shaft/body ossifies by **intramembranous ossification** (directly from mesenchyme, no cartilage model), while the **medial (sternal) epiphysis** ossifies by **endochondral ossification** (via a cartilage model, like a typical long bone) and is notably the **last epiphysis in the entire body to fuse** (~age 25, sometimes later). Do NOT say the clavicle ossifies purely intramembranously — that is a common but incomplete/incorrect statement. The dual mechanism is exactly what makes it unique.

- **Head** (sternal/medial end): rounded, articulates with the manubrium's clavicular notch at the **sternoclavicular joint**.
- **Body/shaft**: transitions from tubular (medial 2/3) to flattened (lateral 1/3); the shape-shift point is marked by the **conoid tubercle** inferiorly.
- **Tail** (acromial/lateral end): flat, blade-like, articulates with the acromion at the **acromioclavicular joint**.
- **Subclavian groove**: inferior surface, middle third — attachment for subclavius muscle.
- **Conoid tubercle** and **trapezoid line**: inferior surface, lateral third — attachment for the coracoclavicular ligament complex (conoid + trapezoid ligaments).

EXAM HACK

Clavicle facts examiners love: (1) first bone to begin ossifying (week 5–6) yet last to finish fusing (~25yo); (2) most commonly fractured bone in the body; (3) fractures classically at the **junction of the middle and lateral thirds** — the weakest point where curvature changes and the cross-section thins; (4) subclavian vessels and brachial plexus lie just deep to the middle third, so severely displaced fractures risk neurovascular injury; (5) it is the **only long bone that ossifies from two primary centres**; (6) it varies with build and sex — thicker/more curved in manual workers, shorter/thinner/less curved in females, and the right clavicle is usually stronger and shorter than the left.

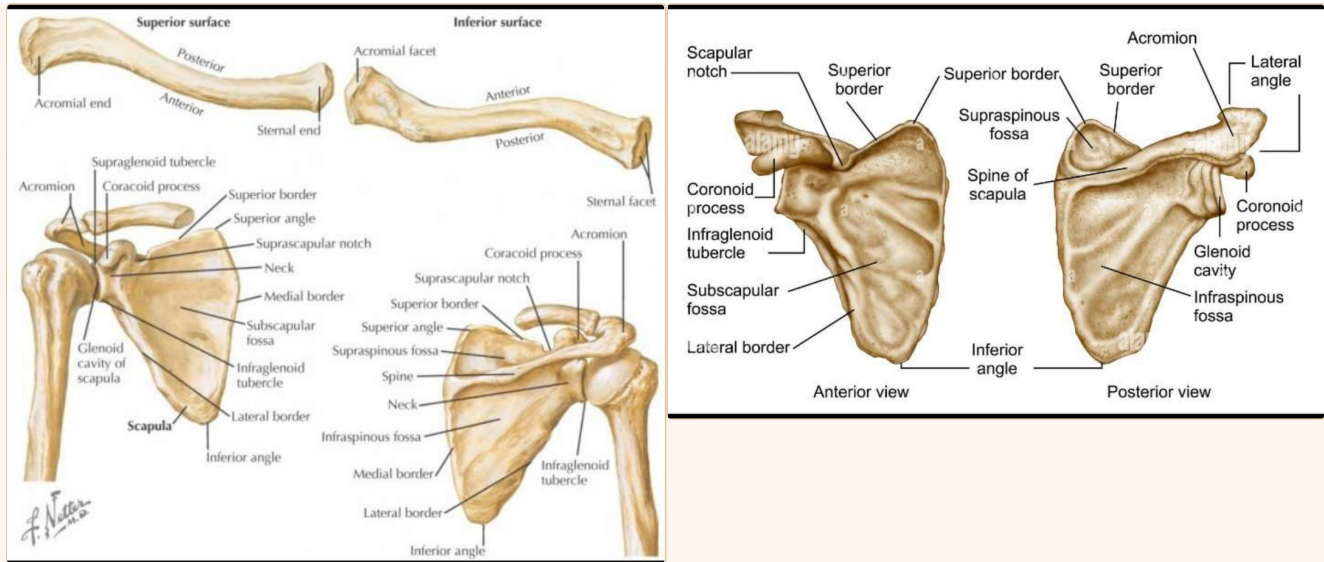
CLINICAL

Falls onto the shoulder (not just onto an outstretched hand) commonly fracture the clavicle. Birth-related clavicle fractures can occur during difficult vaginal deliveries with shoulder dystocia — generally benign and heal well in newborns. **Clavicular dysostosis** — the two primary ossification centres fail to unite, leaving medial and lateral parts separate. **Cleidocranial dysostosis** — partial/complete absence of the clavicle with defective skull ossification. A pierced intermediate supraclavicular nerve can occasionally run through the bone itself.

Scapula

Alzheimer's Mode: A flat triangle glued to the back of the ribcage (overlying ribs 2–7), held on by muscle alone — no direct bony joint to the thorax. Remember it as "3-3-2-3": 3 borders, 3 angles, 2 surfaces, 3 processes.

Feature	Details
3 Borders	Superior (shortest, sharpest — has suprascapular notch), Medial/vertebral (longest), Lateral/axillary (thickest)
3 Angles	Superior angle (~2nd rib level), Inferior angle (~7th rib level — landmark for counting ribs/vertebral levels), Lateral angle (thickest — bears glenoid cavity)
2 Surfaces	Costal/anterior (concave, subscapular fossa — houses subscapularis), Posterior (divided by the spine into supraspinous and infraspinous fossae)
3 Processes	Spine (continues as acromion), Coracoid process (hook-shaped, palpable below clavicle), Glenoid cavity (shallow socket bordered by supraglenoid tubercle above and infraglenoid tubercle below)



Left: clavicle (superior/inferior surfaces) and scapula (glenoid, borders, processes). Right: scapula anterior and posterior views.

- **Suprascapular notch:** superior border, medial to the coracoid process – transmits the **suprascapular NERVE** only (passes beneath the superior transverse scapular ligament).
- **Greater scapular notch:** superior border, lateral to the coracoid (at the base of the spine) – transmits the **suprascapular ARTERY AND NERVE together** from the supraspinous to the infraspinous fossa.

EXAM HACK

"Suprascapular NERVE gets NOTCHED" – the suprascapular nerve alone passes through the **suprascapular** notch. The artery joins the nerve only at the **greater** scapular notch. Also remember "Army over Navy" – the suprascapular Artery passes OVER the superior transverse scapular ligament while the Nerve passes UNDER it, through the suprascapular notch – a classic entrapment site (suprascapular nerve entrapment syndrome).

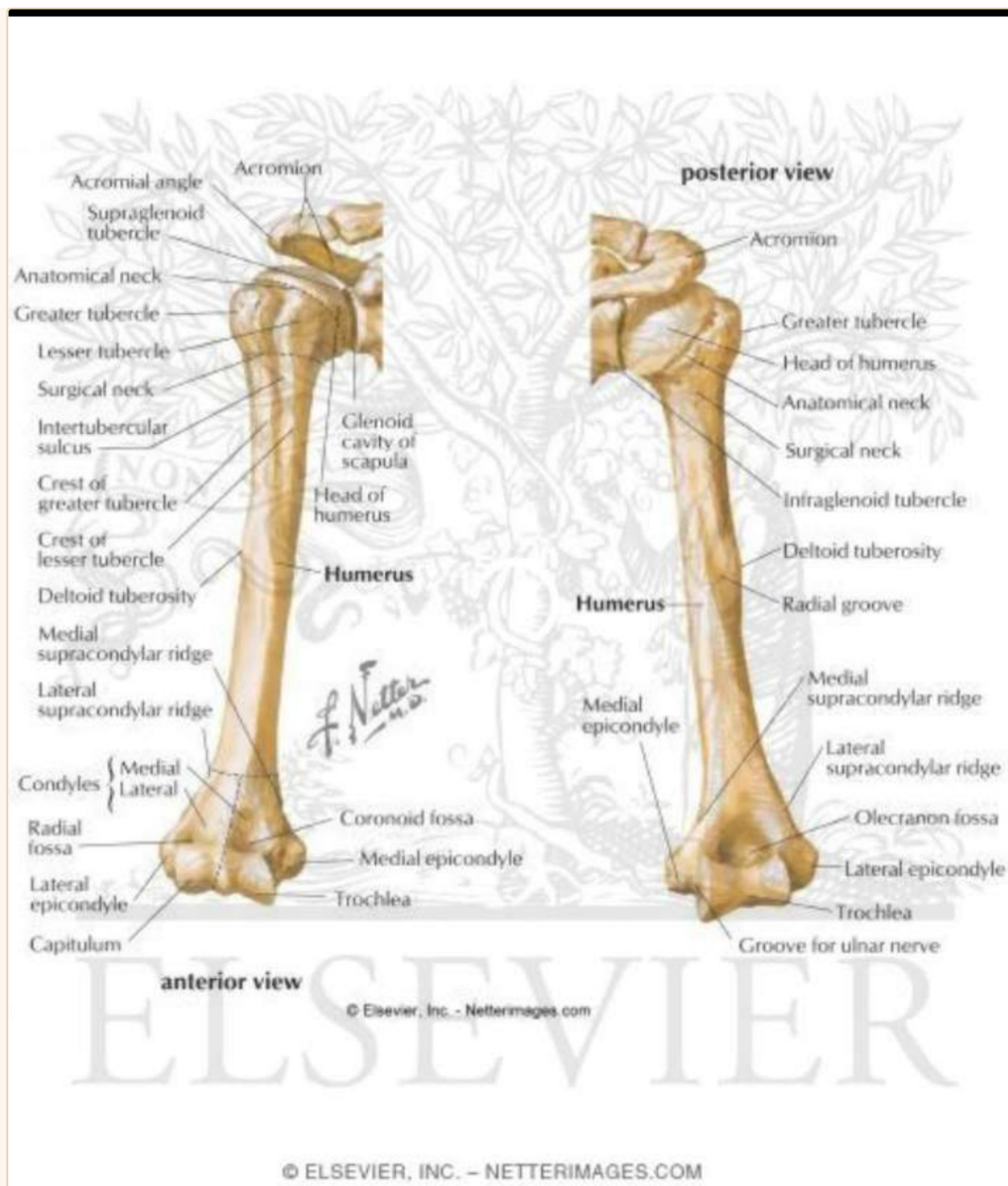
CLINICAL

Winging of the scapula results from long thoracic nerve injury (serratus anterior paralysis) – tested with the wall push-up test. Scapular fractures are uncommon and signal high-energy trauma given the bone's mobility and muscular protection.

Humerus

Alzheimer's Mode: The humerus is "the story of three landmarks that trap three nerves." Learn where each nerve runs relative to the bone and you can answer almost any clinical humerus question. The longest and strongest bone of the upper limb.

Region	Landmark
Proximal	Head (articulates with glenoid) · Anatomical neck · Greater tubercle (SIT muscles insert) · Lesser tubercle (subscapularis inserts) · Intertubercular/bicipital groove (long head of biceps tendon, held by the transverse humeral ligament) · Surgical neck
Shaft	Deltoid tuberosity (lateral, mid-shaft) · Radial/spiral groove (posterior, below deltoid tuberosity)
Distal	Capitulum (lateral, meets radius) · Trochlea (medial, meets ulna) · Medial epicondyle (ulnar nerve groove behind it) · Lateral epicondyle (common extensor origin) · Olecranon fossa (posterior) · Coronoid fossa & radial fossa (anterior)



Humerus, anterior and posterior views — proximal, shaft and distal landmarks labelled.

EXAM HACK — THE 3-NERVE RULE (own this cold)

Site	Nerve at risk
Surgical neck	Axillary nerve (+ posterior circumflex humeral artery)
Radial/spiral groove (mid-shaft)	Radial nerve (+ profunda brachii artery)
Medial epicondyle (groove behind it)	Ulnar nerve — the "funny bone" spot

This single table answers a huge share of upper limb clinical questions. Don't mix up surgical neck (axillary) with radial groove (radial) — they are the two most confused sites.

CLINICAL

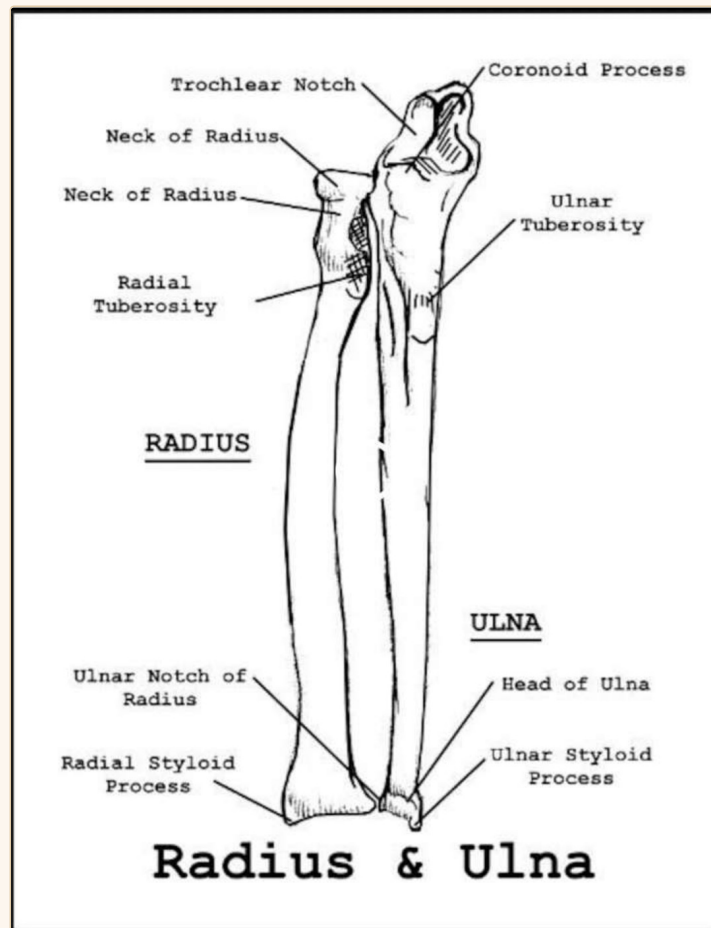
Surgical neck fracture → axillary nerve injury → deltoid paralysis + "regimental badge" sensory loss. Mid-shaft fracture → radial nerve injury → wrist drop. Supracondylar fracture (common in children) → risk to the brachial artery and median nerve. Lateral epicondylitis = "tennis elbow" (common extensor origin); medial epicondylitis = "golfer's elbow" (common flexor origin). An avulsion fracture of the greater tubercle is seen most often in middle-aged and elderly people.

Radius and Ulna

Alzheimer's Mode: "ROLU, ULR" — Radius is On the Lateral side (thumb side) in anatomical position; Ulna is Medial. The ulna is the fixed pivot; the radius rotates around it like a dial spinning around a fixed post. Together they're homologous to the leg's tibia (ulna) and fibula (radius)

pairing... but flipped: radius = mobile (like the fibula is thin/lateral), ulna = stabilizing (like the tibia is weight-bearing/medial).

Ulna (medial, "stable")	Radius (lateral, "mobile")
Olecranon process (point of elbow, triceps insertion) + coronoid process form the C-shaped trochlear notch	Head (disc-shaped, articulates with capitulum + radial notch of ulna)
Radial notch (lateral side of coronoid process — proximal radioulnar joint)	Radial tuberosity (biceps brachii inserts here)
Ulnar tuberosity (below coronoid — muscle attachment)	Oblique line (descends from tuberosity — muscle attachment)
Supinator crest (lateral shaft — muscle attachment)	Shaft widens progressively distally (opposite of ulna)
Head + styloid process (distal end, small) — does NOT articulate directly with carpal bones (separated by the articular disc/TFCC)	Ulnar notch (distal radioulnar joint) + wide distal end articulating directly with scaphoid + lunate



Both bones are connected proximally (proximal radioulnar joint) and distally (distal radioulnar joint), plus along their shafts by the **interosseous membrane**, which stabilizes the two bones during pronation/supination.

EXAM HACK — FOREARM FRACTURE TRIO

Colles' fracture = distal radius fracture, dorsal displacement (FOOSH injury, elderly) → "dinner fork" deformity. **Smith's fracture** = distal radius fracture, volar/anterior displacement (reverse Colles', "garden spade" deformity). **Monteggia fracture** = proximal ulna fracture + radial head dislocation ("Monteggia Ulna, proximal"). **Galeazzi fracture** = distal radius fracture + distal radioulnar joint dislocation ("Galeazzi Radius, distal"). **Nightstick fracture** = isolated ulnar shaft fracture from a direct blow (classic defensive injury). Mnemonic: "**MUGR**" — Monteggia = Ulna (proximal); Galeazzi = Radius (distal).

CLINICAL

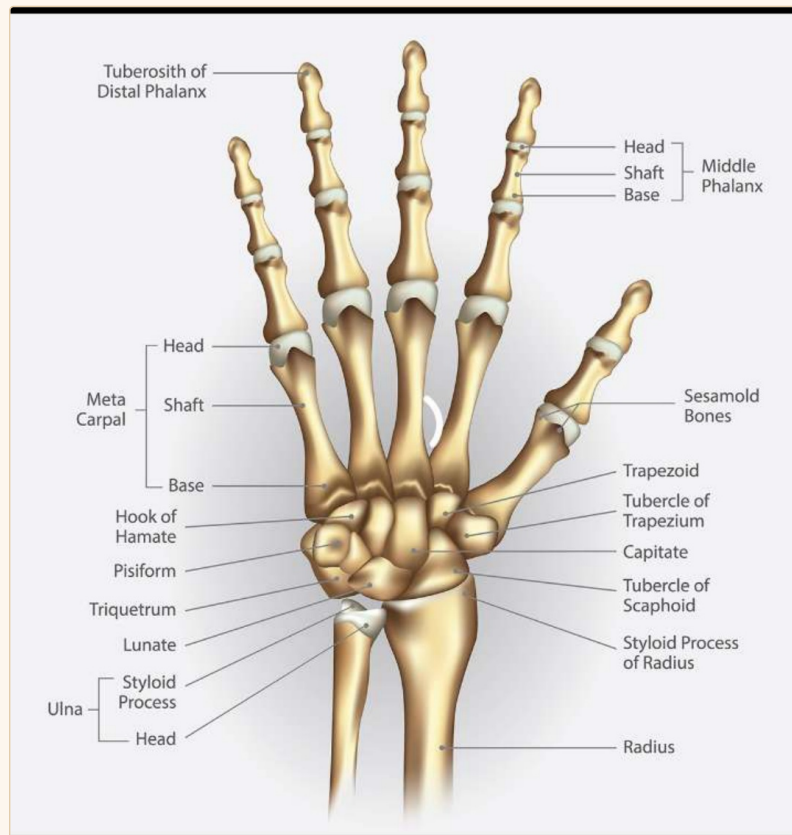
The ulna's lack of direct wrist articulation (separated by the fibrocartilaginous disc/TFCC) means the radius transmits the vast majority of force from hand to forearm — this is exactly why distal radius fractures (Colles', Smith's) are so common and clinically significant. Colles' fracture is the most common forearm fracture, typically in adults over 50 (especially women with osteoporosis) after a FOOSH.

Bones of the Wrist and Hand

Alzheimer's Mode: 8 carpal bones in two rows of 4, like a "roundabout of small stones" between the forearm and the metacarpals.

Proximal row (lateral → medial): Scaphoid (articulates with radius) · Lunate (articulates with radius) · Triquetrum (articulates with the distal row) · Pisiform (sesamoid bone within flexor carpi ulnaris tendon, sits atop triquetrum — the bulging bone of the wrist).

Distal row (lateral → medial): Trapezium (with its tubercle — articulates with scaphoid + thumb metacarpal, forms the saddle joint) · Trapezoid (articulates with index metacarpal) · Capitate (largest carpal bone, articulates with middle metacarpal) · Hamate (with its hook — articulates with ring + little metacarpals).



EXAM HACK

Mnemonic (proximal row → distal row, lateral → medial): **"Some Lovers Try Positions That They Cannot Handle"** — Scaphoid, Lunate, Triquetrum, Pisiform, Trapezium, Trapezoid, Capitate, Hamate. **Scaphoid** = most commonly fractured carpal bone; blood supply enters distally and runs **retrograde**, so fractures near the proximal pole risk avascular necrosis (same theme as the femoral neck). **Lunate** = most commonly dislocated carpal bone. **Pisiform** = smallest carpal, a true sesamoid bone. **Capitate** = largest carpal bone.

Each metacarpal (5 total, numbered I–V thumb→little finger) has a base, shaft, and head. The 1st metacarpal forms a **saddle joint** with the trapezium, enabling opposition. Phalanges: thumb has 2 (no middle phalanx), fingers 2–5 each have 3 (proximal, middle, distal) — 14 phalanges total.

CLINICAL

Boxer's fracture = neck of the 5th metacarpal (from punching). **Bennett's fracture** = fracture-dislocation at the base of the 1st metacarpal. **Hook of hamate fracture** = seen in golfers/racket-sport athletes from repetitive gripping stress. Snuffbox tenderness after a fall → always suspect scaphoid fracture until proven otherwise.

3. JOINTS AND LIGAMENTS OF THE UPPER LIMB

Sternoclavicular (SC) Joint

Alzheimer's Mode: The only bony connection between the entire upper limb and the axial skeleton — a saddle-type synovial joint reinforced by 4 ligaments, with an articular disc for shock absorption, and remarkably strong for its size.

- **Anterior sternoclavicular ligament:** clavicular head → anterior clavicular notch of manubrium.
- **Posterior sternoclavicular ligament:** clavicular head → posterior clavicular notch.
- **Costoclavicular ligament:** inferior proximal clavicle → 1st rib + its costal cartilage.
- **Interclavicular ligament:** connects both clavicular heads across the top of the manubrium.

Other Clavicular Ligaments (Coracoclavicular Complex + AC Ligament)

- **Conoid ligament:** conoid tubercle (clavicle) → base of coracoid process.
- **Trapezoid ligament:** trapezoid line (clavicle) → superior surface of coracoid process.
- Conoid + Trapezoid together = the **coracoclavicular ligament** — the primary stabilizer of the distal clavicle/AC joint.
- **Acromioclavicular ligament:** reinforces the AC joint capsule directly (distal clavicle ↔ acromion).

EXAM HACK

AC joint separations are graded by which ligaments tear: Grade I = AC ligament sprain only; Grade III+ = both AC AND coracoclavicular ligaments torn, causing visible step deformity. The coracoclavicular ligament is the TRUE stabilizer — its integrity is what surgeons check on stress X-rays. A fall onto the point of the shoulder is the classic mechanism for AC joint injury in contact sports.

Glenohumeral (Shoulder) Joint

Alzheimer's Mode: A golf ball (humeral head) sitting on a golf tee (shallow glenoid). Greatest range of motion of any joint in the body — at the cost of being the most frequently dislocated joint. Stability here is almost entirely soft-tissue, the exact opposite of the hip.

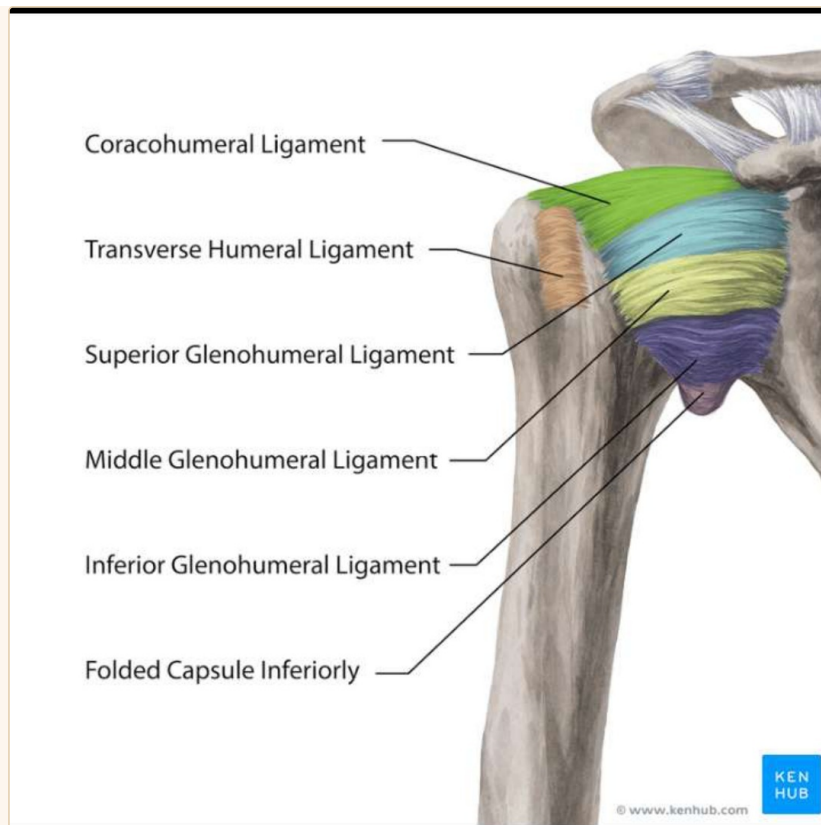
Type: Synovial ball-and-socket joint — the most mobile joint in the body, but the least bony-stable.

Static Stabilizers

- **Glenoid labrum:** fibrocartilaginous rim, deepens the socket.
- **Joint capsule:** deliberately loose/lax to allow the huge range of motion.
- **Glenohumeral ligaments** (capsule thickenings): Superior GHL (below supraglenoid tubercle → superomedial greater tubercle), Middle GHL (anterior glenoid rim → mid-anatomical neck), Inferior GHL (inferior glenoid rim → inferior surgical neck) — the **IGHL is the most important, acting like a hammock, and is the primary restraint against anterior dislocation**.
- **Coracohumeral ligament:** coracoid process → greater + lesser tubercles; supports the weight of the arm.
- **Coracoacromial ligament:** coracoid process → acromion; forms the coracoacromial arch (a protective roof), NOT a true joint ligament (no humeral attachment).
- **Transverse humeral ligament:** bridges the intertubercular groove, holding the long head of biceps tendon in place.

Dynamic Stabilizer

The **rotator cuff** (SITS muscles) actively holds the humeral head centered in the glenoid — the shoulder's stability is muscle-dominant, unlike the hip's ligament-dominant design.



Glenohumeral ligaments and the coracoacromial arch, anterior view.

Bursae of the Glenohumeral Joint

Bursa	Location	Note
Subacromial (subdeltoid)	Below acromion/deltoid, above supraspinatus tendon	Most clinically important — subacromial bursitis + impingement occur here together
Subcoracoid	Between coracoid process and middle GHL	Often continuous with subacromial bursa
Subscapular	Below subscapularis tendon, anterosuperior scapula	Usually communicates with the joint cavity directly

EXAM HACK

Static vs Dynamic stabilizers is THE defining shoulder concept examiners test. The coracoacromial arch (coracoacromial ligament + acromion + coracoid process) forms a rigid roof over supraspinatus — repetitive overhead movement pinches the tendon here → subacromial impingement syndrome, often WITH subacromial bursitis since they share the same tight space.

CLINICAL

Anterior shoulder dislocation (~95% of all shoulder dislocations) tears the IGL/anteroinferior labrum → **Bankart lesion**. The humeral head impacts the anterior glenoid rim → **Hill-Sachs lesion** (posterolateral humeral head compression fracture) — the two are commonly found together. Always test sensation over the "regimental badge" area (axillary nerve) after any shoulder dislocation. **Adhesive capsulitis (frozen shoulder)**: capsule thickens/contracts, common in diabetics. **Painful arc** (60–120° abduction) = classic sign of subacromial impingement.

Elbow Joint

Alzheimer's Mode: A pure hinge joint — flexion/extension only, no rotation (rotation happens at the separate radioulnar joints, not here).

Type: Synovial hinge joint — humero-ulnar (trochlea ↔ trochlear notch, the main hinge) + humero-radial (capitulum ↔ radial head) articulations.

- **Radial collateral ligament:** lateral epicondyle → anterolateral radius (blends with annular ligament).
- **Ulnar collateral ligament:** medial epicondyle → proximal ulna (anterior band is the primary stabilizer — the one reconstructed in "Tommy John surgery").
- **Annular ligament:** wraps around the radial head, anchoring it against the ulna's radial notch — permits pronation/supination while keeping the radial head in place.

CLINICAL

Supracondylar humeral fracture (children) risks the brachial artery and median nerve, which lie just anterior to it. "Pulled elbow" (radial head subluxation) in toddlers occurs when the annular ligament slips over the radial head from a sudden pull on the arm. Ulnar collateral ligament injury is common in throwing athletes.

Wrist (Radiocarpal) Joint

Alzheimer's Mode: Formed between the distal radius and the proximal carpal row (mainly scaphoid + lunate) — the ulna is NOT directly part of this joint, separated by the articular disc.

Type: Condyloid (ellipsoid) synovial joint. Movements: flexion, extension, abduction (radial deviation), adduction (ulnar deviation), circumduction.

- **Radial collateral ligament:** radial styloid process → scaphoid + trapezium; prevents excessive adduction (ulnar deviation is limited).
- **Ulnar collateral ligament:** ulnar styloid process → triquetrum + pisiform; prevents excessive abduction (radial deviation is limited).
- **Palmar radiocarpal ligament:** palmar distal radius → carpal bones; aids supination.
- **Dorsal radiocarpal ligament:** dorsal radius → carpal bones/thumb-index metacarpals; aids pronation.
- **Palmar ulnocarpal ligament:** ulnar styloid → lunate + triquetrum.

EXAM HACK

Don't confuse the naming direction with the movement it resists — radial collateral ligament resists ulnar deviation (adduction of the hand), NOT radial deviation. It's named for its location, not the movement it allows.

Carpometacarpal (CMC) and Metacarpophalangeal (MCP) Joints

- **CMC joints:** distal carpal row ↔ metacarpal bases. The 1st CMC (thumb) is a unique saddle joint, allowing opposition; the others are largely plane/gliding joints.
- **MCP joints** (the "knuckles"): condyloid joints, reinforced by the circumferential volar (palmar) plate, collateral ligaments on each side, and 3 deep transverse metacarpal ligaments (linking the index, middle, and ring MCP joints together, limiting their spread).

Interphalangeal (IP) Joints

Pure hinge joints (flexion/extension only) between the phalanges — PIP and DIP in fingers 2-5; the thumb has only a single IP joint (no middle phalanx).

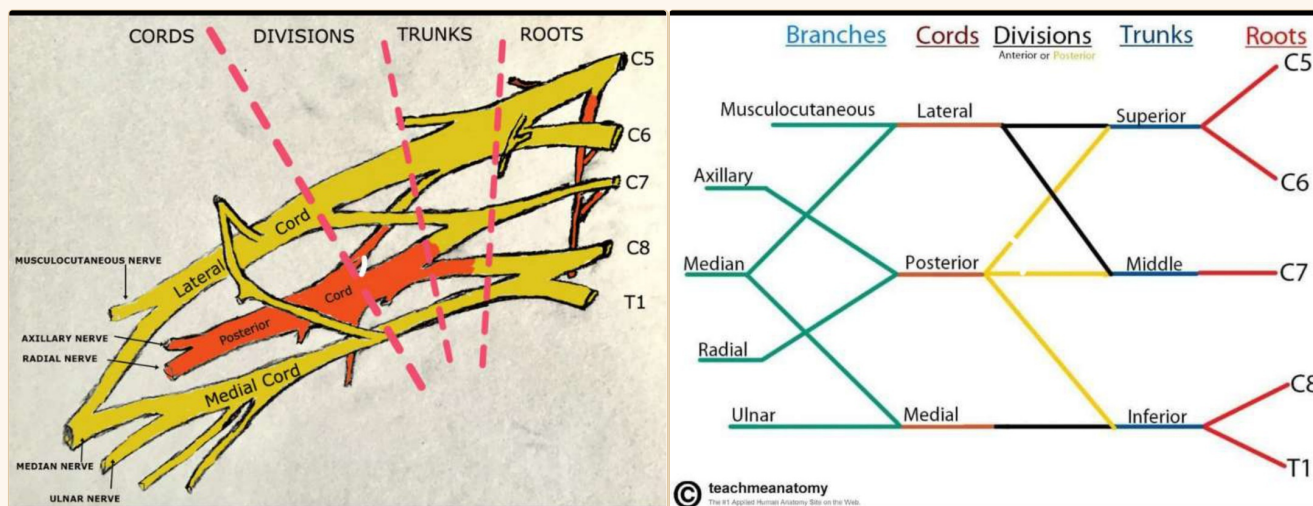
EXAM HACK — JOINT TYPES SUMMARY

The **glenohumeral joint** is the ONLY true ball-and-socket joint in the upper limb. The **elbow and interphalangeal joints** are both pure hinge joints. The **1st CMC (thumb)** is a unique saddle joint. Everything else (SC, AC, radioulnar, most CMC, MCP) falls somewhere between plane and condyloid — but glenohumeral = ball-and-socket and elbow/IP = hinge are the two facts examiners test most.

4. NERVE SUPPLY — THE BRACHIAL PLEXUS

Alzheimer's Mode: Picture a highway system — 5 roots merge into 3 trunks, each trunk splits into 2 divisions, and the divisions regroup into 3 cords, which finally branch into 5 terminal nerves. Remember it as **"Randy Travis Drinks Cold Beer"** — Roots, Trunks, Divisions, Cords, Branches.

Stage	Number	Key point
Roots	5 (C5–T1)	Anterior rami of C5–T1; lie between scalenus anterior & medius
Trunks	3	Superior (C5,C6), Middle (C7), Inferior (C8,T1); found in the posterior triangle of the neck
Divisions	6	Each trunk splits into an anterior and a posterior division behind the clavicle
Cords	3	Lateral, Posterior, Medial — named by relation to the axillary artery, in the axilla
Branches	5 terminal (+ several minor)	Musculocutaneous, Axillary, Radial, Median, Ulnar



Roots → Trunks → Divisions → Cords → Terminal Branches of the brachial plexus.

Cord	Formed from	Gives rise to
Lateral cord	Anterior divisions of superior + middle trunks (C5,6,7)	Musculocutaneous nerve (alone) + lateral head of median nerve
Posterior cord	Posterior divisions of ALL THREE trunks (C5–T1) — the only cord touched by every root	Axillary nerve + Radial nerve
Medial cord	Anterior division of the inferior trunk only (C8,T1)	Ulnar nerve (alone) + medial head of median nerve

EXAM HACK

Mnemonic for the 5 terminal branches: **"Matt Ate Rats And Mice"** — Musculocutaneous, Axillary, Radial, (median straddles both), Ulnar. The **median nerve is the only nerve formed from TWO cords** (lateral + medial), giving it its characteristic "M" shape straddling the axillary artery. **Posterior cord = only cord touching every root (C5–T1)** because it collects the posterior division of all 3 trunks — this is why posterior cord injury has the broadest deficit of the three cords.

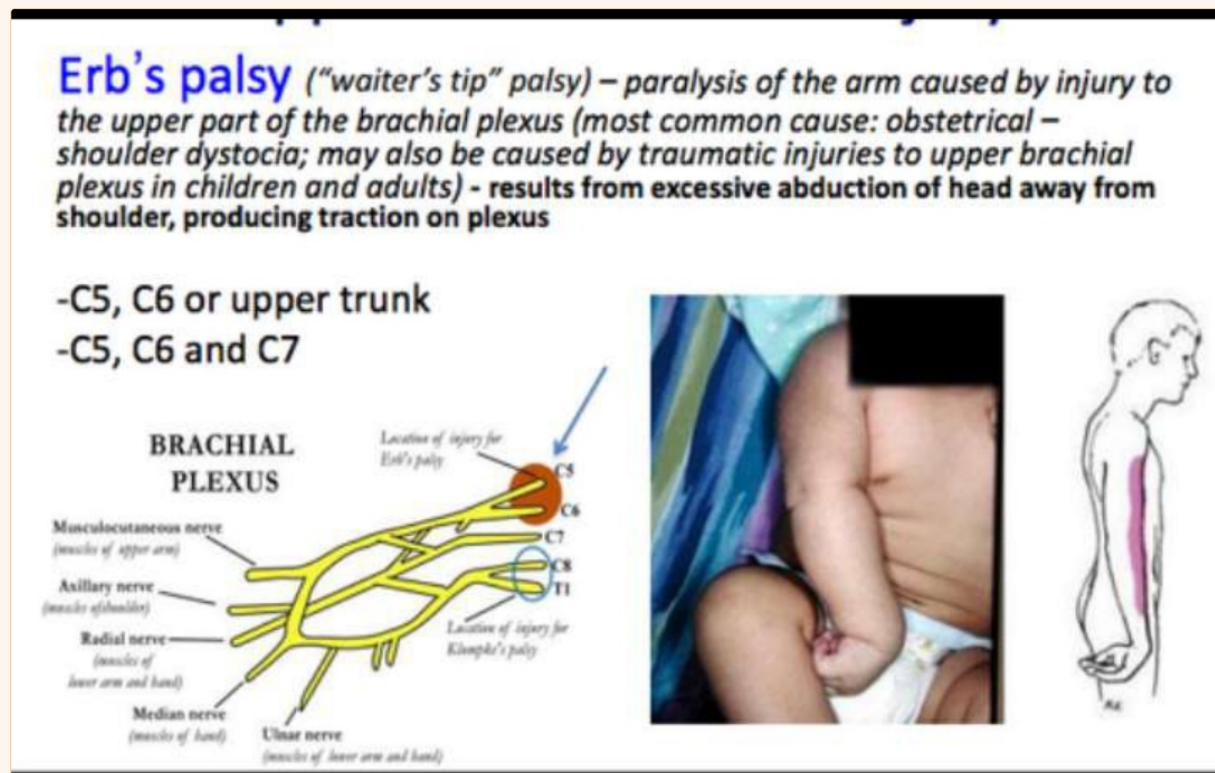
EXAM HACK — PRE-TRUNK (ROOT-LEVEL) BRANCHES

Some nerves branch directly off the **ROOTS**, before the trunks even form — a classic trick question lists these among "trunk branches" when they're not. Includes: **Long thoracic nerve** (C5,6,7 → serratus anterior), **Dorsal scapular nerve** (C5 → rhomboids + levator scapulae), **Nerve to subclavius** (C5,C6 → subclavius). Also root/trunk-level: **Suprascapular nerve** branches off the **superior trunk** (not a root) → supraspinatus + infraspinatus.

Terminal Branches at a Glance

Nerve	Cord	Roots	Main action
Musculocutaneous	Lateral	C5–C7	Flexes elbow (biceps, brachialis, coracobrachialis); sensory to lateral forearm
Median	Lateral + Medial	C5–T1	Flexes wrist/most digits, thumb opposition; "hand of benediction" if injured
Ulnar	Medial	C8,T1	1½ forearm flexors + intrinsic hand muscles; claw hand if injured

Axillary	Posterior	C5,C6	Deltoid, teres minor; sensory over the lateral shoulder ("regimental badge")
Radial	Posterior	C5-T1	Thickest branch; extends elbow/wrist/fingers ("BEST" – Brachioradialis, Extensors, Supinator, Triceps); wrist drop if injured



Erb's palsy – upper plexus injury (C5,C6) producing the classic "waiter's tip" posture.

CLINICAL – ERB'S vs KLUMPKE'S PALSY

Erb's palsy (upper trunk, C5–C6): birth trauma or motorcycle accident forcing the head away from the shoulder → **"waiter's tip" position** (arm adducted, medially rotated, elbow extended, forearm pronated, wrist flexed) – deltoid, supraspinatus, infraspinatus, biceps, brachialis all weak. "Erb's = Elevated/upper trunk = can't lift the arm."

Klumpke's palsy (lower trunk, C8–T1): arm pulled upward forcefully (breech delivery, grabbing something while falling) → **claw hand** (loss of intrinsic hand muscles) ± **Horner's syndrome** if T1 root is avulsed (ptosis, miosis, anhidrosis – sympathetic fibers travel with T1). "Klumpke's = can't use the hand."

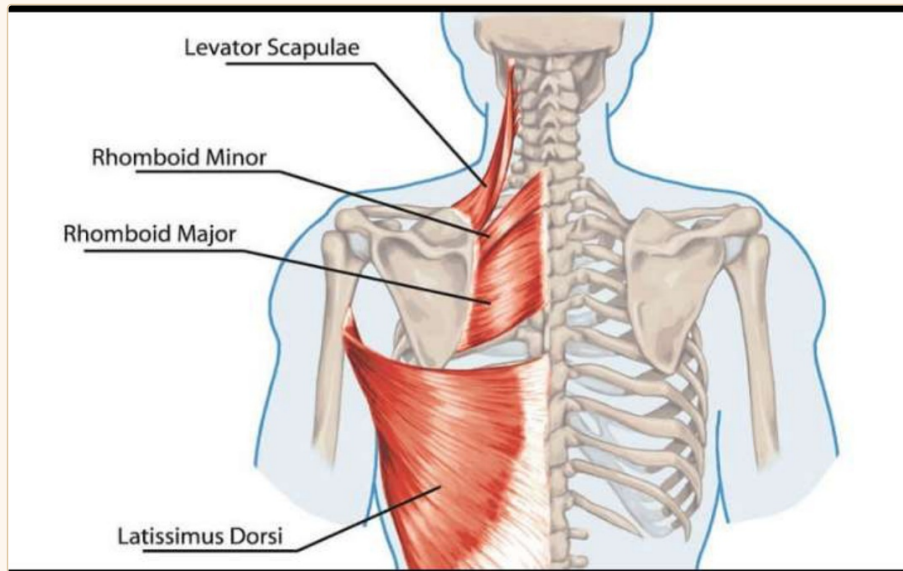
Long thoracic nerve injury → winged scapula (serratus anterior palsy). **Winging tested with the wall push-up test.** A prefixed plexus draws heavily on C4 with little/no T2 contribution; a postfixed plexus draws more on T2 with a smaller C5 – a one-segment shift worth recognising if a question describes "unusual" root involvement.

5. MUSCULATURE OF THE UPPER LIMB

Upper limb muscles are grouped by region: scapular stabilizers (back), shoulder-joint movers (rotator cuff + deltoid + teres major), thoracic wall muscles, the arm, the forearm, and the intrinsic hand muscles.

Muscles Arising from the Back (Scapular Stabilizers)

Alzheimer's Mode: These are the "puppet strings" of the scapula — since the scapula has no direct bony joint to the ribcage, these muscles anchor and reposition it, giving the rotator cuff a stable base to work from.



Muscle	Origin	Insertion	Action	Nerve
Trapezius	Superior nuchal line, occipital protuberance, nuchal ligament, spinous processes C7-T12	Lateral clavicle, acromion, scapular spine	Elevates (upper), retracts (middle), depresses (lower) scapula; all parts rotate scapula upward	Accessory nerve (CN XI) + C3,C4 proprioceptive
Levator scapulae	Posterior tubercles, transverse processes C1-C4	Superior angle → medial border of scapula	Elevates scapula	C3, C4 + dorsal scapular nerve
Rhomboid minor	Nuchal ligament + spinous processes C7-T1	Medial border, at level of scapular spine	Retracts, stabilizes, assists elevation/rotation	Dorsal scapular nerve
Rhomboid major	Spinous processes T2-T5	Medial border, below the spine (below rhomboid minor)	Retracts, stabilizes scapula	Dorsal scapular nerve
Serratus anterior	Outer surfaces of ribs 1-8/9	Costal (anterior) surface of the medial border of scapula	Protracts scapula; rotates it upward for full overhead abduction	Long thoracic nerve
Latissimus dorsi	Spinous processes T6-L2, ribs 8-12, thoracolumbar fascia, iliac crest	Floor of intertubercular groove of humerus	Adduction, extension, internal rotation of the arm	Thoracodorsal nerve

EXAM HACK

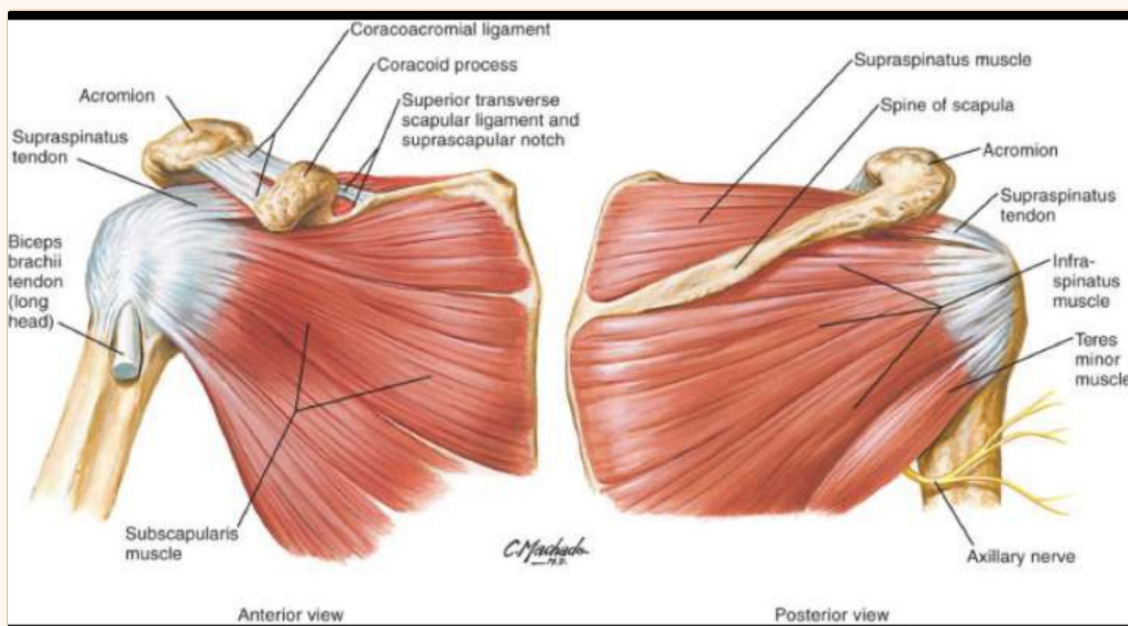
Trapezius is the **ONLY** muscle here supplied by a **cranial nerve** (CN XI) rather than a plexus branch — a classic "odd one out" trap. Levator scapulae + both rhomboids share the same **dorsal scapular nerve** (a pre-trunk root-level branch of the brachial plexus, arising directly from C5, before the trunks even form — same pattern as the long thoracic nerve). Rhomboid minor sits **ABOVE** rhomboid major (minor = C7-T1, major = T2-T5, lower). **Serratus anterior + trapezius = the force couple for full overhead arm raising** — if either is weak, full abduction fails even with a perfect rotator cuff. Latissimus dorsi's insertion completes a 3-muscle trio at the intertubercular groove: **pectoralis major (lateral lip), latissimus dorsi (floor), teres major (medial lip)** — "a Lady between two Majors."

CLINICAL

Long thoracic nerve injury → winged scapula (serratus anterior palsy), tested with the wall push-up test. Accessory nerve injury (e.g., neck lymph node biopsy) → trapezius weakness, shoulder droop, difficulty raising the arm above horizontal. Dorsal scapular nerve injury → subtle winging + weak retraction. Latissimus dorsi flap is used surgically in breast reconstruction; thoracodorsal nerve injury (e.g., during axillary lymph node dissection) weakens climbing/pulling movements.

Muscles Arising from the Scapula (Rotator Cuff + Shoulder Movers)

Alzheimer's Mode: The rotator cuff is 4 muscles acting like guy-wires holding the humeral head steady in the shallow glenoid "saucer" — without them, every arm-raise would risk dislocation.



Muscle	Origin	Insertion	Action	Nerve
Supraspinatus (SITS)	Supraspinous fossa	Greater tubercle (top facet)	Initiates first 15° of abduction	Suprascapular nerve
Infraspinatus (SITS)	Infraspinous fossa	Greater tubercle (middle facet)	Lateral (external) rotation	Suprascapular nerve
Teres minor (SITS)	Lateral border of scapula	Greater tubercle (lower facet)	Lateral rotation	Axillary nerve
Subscapularis (SITS)	Subscapular fossa	Lesser tubercle	Medial (internal) rotation	Upper + lower subscapular nerves
Deltoid	Scapular spine, acromion, lateral clavicle	Deltoid tuberosity	Abduction (15-90°, middle fibers), flexion (anterior fibers), extension (posterior fibers)	Axillary nerve
Teres major	Inferior angle of scapula (dorsal surface)	Medial lip of intertubercular groove	Extension, adduction, internal rotation	Lower subscapular nerve

EXAM HACK — SITS = ROTATOR CUFF

Supraspinatus, Infraspinatus, Teres minor, Subscapularis. Three of the four (S, I, T-minor) insert on the **greater tubercle** and — except teres minor (axillary nerve) — are **suprascapular nerve**. Subscapularis is the odd one out: different tubercle (lesser), different action (medial not lateral rotation). **Abduction relay:** 0-15° = supraspinatus → 15-90° = deltoid → beyond 90° = trapezius + serratus anterior rotating the scapula (scapulohumeral rhythm, ~2:1 glenohumeral:scapulohumeral ratio). Teres major is NOT part of the rotator cuff despite the similar name to teres minor — easy trap.

CLINICAL

Supraspinatus is the most commonly torn rotator cuff tendon (repetitive overhead strain, impingement under the coracoacromial arch). Axillary nerve injury (surgical neck fracture, anterior dislocation) → deltoid + teres minor paralysis, flattened "squared-off" shoulder, regimental badge sensory loss.

Muscles Arising from the Thoracic Wall

Muscle	Origin	Insertion	Action	Nerve
Pectoralis major	Clavicular head: medial clavicle. Sternocostal head: sternum + costal cartilages ribs 1-6	Lateral lip of intertubercular groove	Adduction, flexion, internal rotation of the arm (glenohumeral joint)	Medial + lateral pectoral nerves (C5-T1)
Pectoralis	Anterior surface, ribs 3-5	Coracoid	Depresses/protracts the scapula ; elevates ribs	Medial pectoral

minor		process	in forced inspiration	nerve (C8,T1)
Subclavius	1st rib + costal cartilage	Subclavian groove of clavicle	Depresses the clavicle (and thus the shoulder); cushions the neurovascular bundle beneath clavicle	Nerve to subclavius (C5,C6)

CORRECTION — DON'T LUMP THESE THREE TOGETHER

Pectoralis major acts on the ARM at the glenohumeral joint — it does NOT move the scapula. Pectoralis minor and subclavius act on the SCAPULA/CLAVICLE, not the arm. This is a classic exam trap: examiners deliberately group these three by region (thoracic wall) to test whether you know they don't share the same functional target. Pec major belongs functionally with latissimus dorsi and teres major (all move the humerus); pec minor belongs with the scapular stabilizers (trapezius, rhomboids, levator scapulae, serratus anterior).

EXAM HACK

Pec major's two heads have opposite-feeling actions depending on arm position: the clavicular head flexes the arm (front raise); the sternocostal head extends the arm from a flexed position (pulling something down) — a "why does one muscle have two different actions" trap. Pectoralis minor's position defines the **3 parts of the axillary artery** and the **3 axillary lymph node levels** — the single most repeated landmark in upper limb anatomy. Subclavius physically cushions the subclavian vessels/brachial plexus, which is partly why clavicle fractures don't injure them as often as proximity would suggest.

CLINICAL

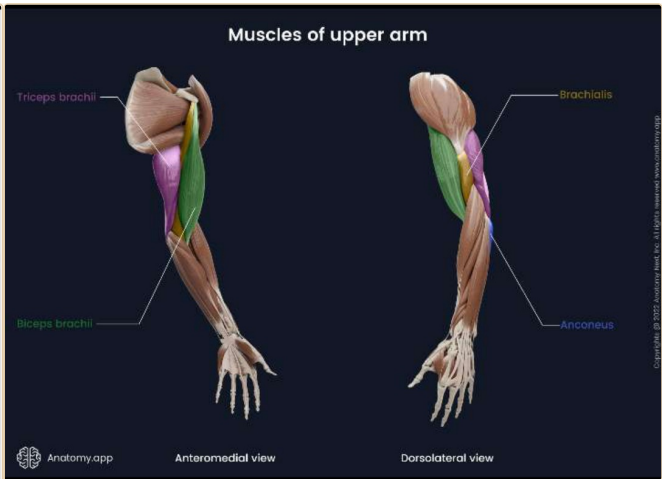
Pectoralis major tear/rupture occurs during heavy bench pressing — sudden pain, bruising, weak adduction/internal rotation. Poland syndrome = congenital absence of the sternocostal head of pec major (classic exam mention). Pectoralis minor syndrome = neurovascular compression beneath a tight pec minor, a thoracic-outlet-type presentation.

Scapular Movements — The Full System

The scapula moves in 6 directions, and virtually every muscle you've just learned contributes to at least one:

Movement	Muscles
Elevation (shrugging)	Trapezius (upper fibers) + Levator scapulae
Depression	Trapezius (lower fibers) + Pectoralis minor + Subclavius (indirect)
Protraction (arm adducted, reaching forward)	Serratus anterior + Pectoralis minor
Retraction (arm abducted)	Trapezius (middle fibers) + Rhomboids
Upward rotation (arms raised overhead)	Trapezius + Serratus anterior (force couple)
Downward rotation (arms by side)	Rhomboids + Levator scapulae + Pectoralis minor

Muscles of the Arm — Anterior Compartment



Muscle	Origin	Insertion	Action	Nerve
Biceps brachii	Long head: supraglenoid tubercle. Short head: coracoid process tip	Radial tuberosity + bicipital aponeurosis	Flexes elbow; powerful supinator (strongest at 90° flexion)	Musculocutaneous nerve
Coracobrachialis	Coracoid process tip	Medial humeral shaft	Flexes + adducts the arm at shoulder	Musculocutaneous nerve
Brachialis	Distal anterior humeral shaft	Ulnar tuberosity	Main flexor of the elbow — works in ALL forearm positions	Musculocutaneous nerve AND radial nerve (dual innervation)

Muscle of the Arm — Posterior Compartment

Muscle	Origin	Insertion	Action	Nerve
Triceps brachii	Long head: infraglenoid tubercle. Lateral head: above the radial groove. Medial head: below the radial groove	Olecranon process	Extends elbow + arm; long head also extends/adducts shoulder	Radial nerve
Anconeus	Lateral epicondyle	Olecranon process	Assists elbow extension; stabilizes ulna during pronation	Radial nerve

EXAM HACK

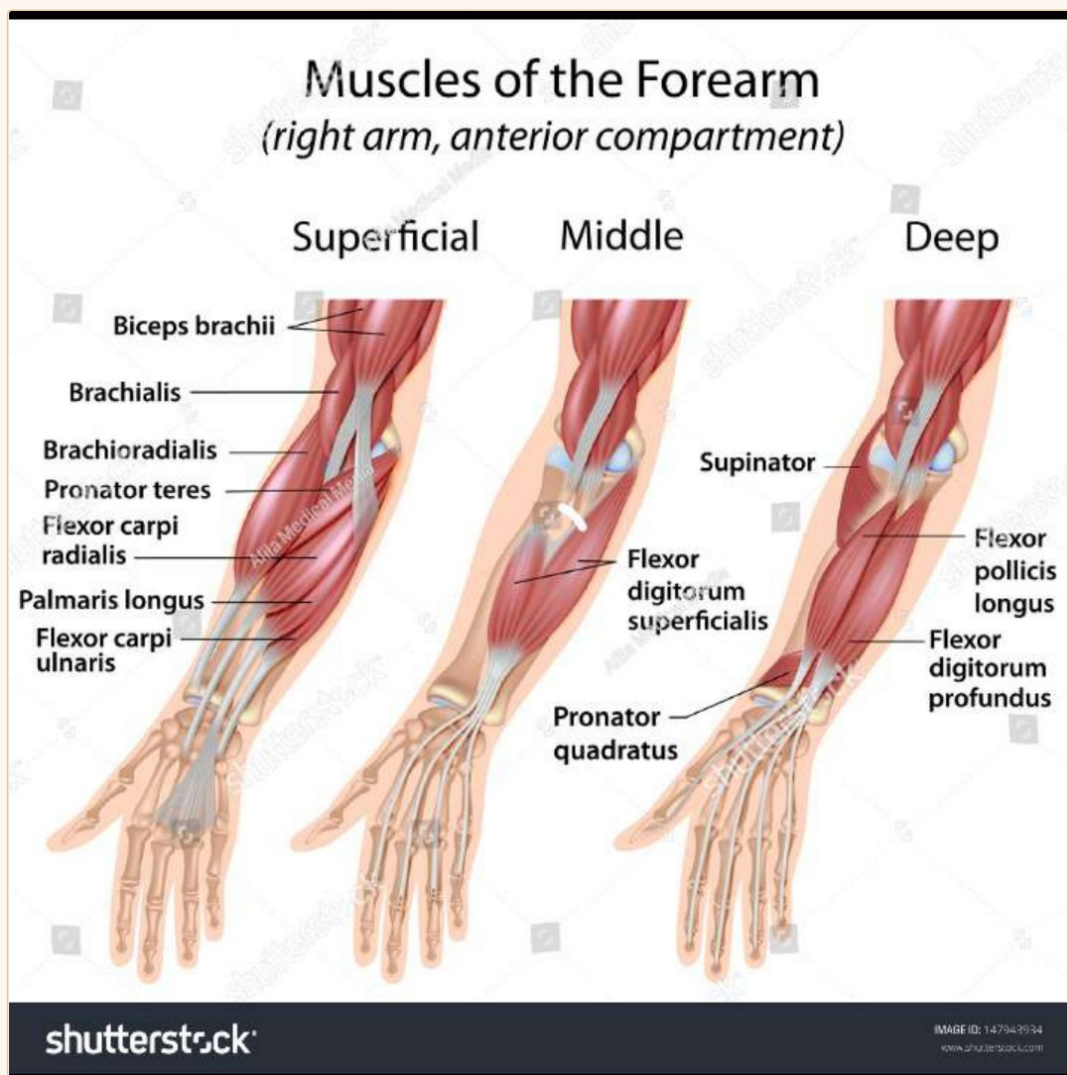
Brachialis is the true prime elbow flexor — not biceps. Biceps is actually a better SUPINATOR than flexor, and only flexes efficiently when the forearm is supinated. Brachialis flexes in every position. **Brachialis is the only anterior-arm muscle with dual innervation** (musculocutaneous + radial) — a classic "odd one out" question, and a pattern that repeats later with flexor digitorum profundus in the forearm. The radial nerve runs in the spiral groove sandwiched between triceps' lateral and medial heads — this is why mid-shaft humeral fractures classically injure the radial nerve.

CLINICAL

Distal biceps tendon rupture — sudden pop with a "Popeye deformity" bulge in the arm, from forceful eccentric loading. Triceps reflex tests the C7 nerve root and radial nerve integrity.

Muscles of the Forearm — Anterior (Flexor) Compartment

Alzheimer's Mode: Almost all originate from ONE spot — the common flexor origin on the medial epicondyle — like ropes tied to a single anchor point. Nerve rule: median nerve runs the show, except ulnar nerve gets exactly 1½ muscles.



Superficial layer

Muscle	Insertion	Action	Nerve
Pronator teres (most lateral)	Lateral radial shaft	Pronation, assists elbow flexion	Median
Flexor carpi radialis	Base of 2nd/3rd metacarpals	Flexion + abduction of wrist	Median
Palmaris longus (variably absent)	Palmar aponeurosis	Flexion of wrist	Median
Flexor carpi ulnaris (most medial)	Pisiform, hamate, 5th metacarpal base	Flexion + adduction of wrist	Ulnar

Intermediate layer

Muscle	Insertion	Action	Nerve
Flexor digitorum superficialis	Middle phalanges of fingers 2-5	Flexes PIP joints (+ wrist)	Median

Deep layer

Muscle	Insertion	Action	Nerve
Flexor digitorum profundus	Distal phalanges of fingers 2-5	Flexes DIP joints	Ulnar (medial half) + Anterior interosseous nerve (lateral half) — split/dual
Flexor pollicis longus	Distal phalanx of thumb	Flexes thumb IP joint	Anterior interosseous nerve (median branch)
Pronator quadratus	Distal radius	Pronation	Anterior interosseous nerve

(deepest)

EXAM HACK — THE "1½ MUSCLE" ULNAR RULE

In the entire anterior forearm, the ulnar nerve supplies ONLY: flexor carpi ulnaris (1 full muscle) + the medial half of flexor digitorum profundus (½ muscle). Everything else is median territory. **FDP mirrors brachialis's dual-innervation trap** — both are the "exception" muscle in their compartment. To test FDS alone clinically: hold the other 3 fingers straight (disables FDP's shared tendon action) and ask the patient to flex the tested finger — if only the PIP flexes, FDS is intact.

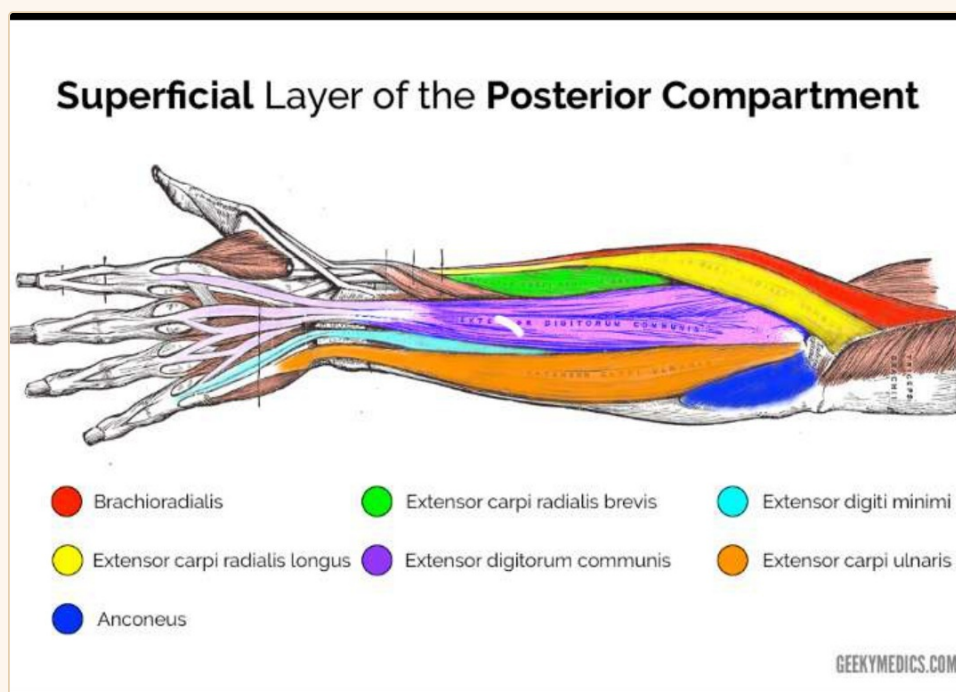
CLINICAL

Carpal tunnel syndrome compresses the median nerve at the wrist — but NOT its territory in the forearm (median nerve branches to forearm flexors leave before the carpal tunnel). Anterior interosseous nerve syndrome selectively weakens FPL, lateral FDP, and pronator quadratus — presents as inability to make an "OK sign" (pinch grip).

Muscles of the Forearm — Posterior (Extensor) Compartment

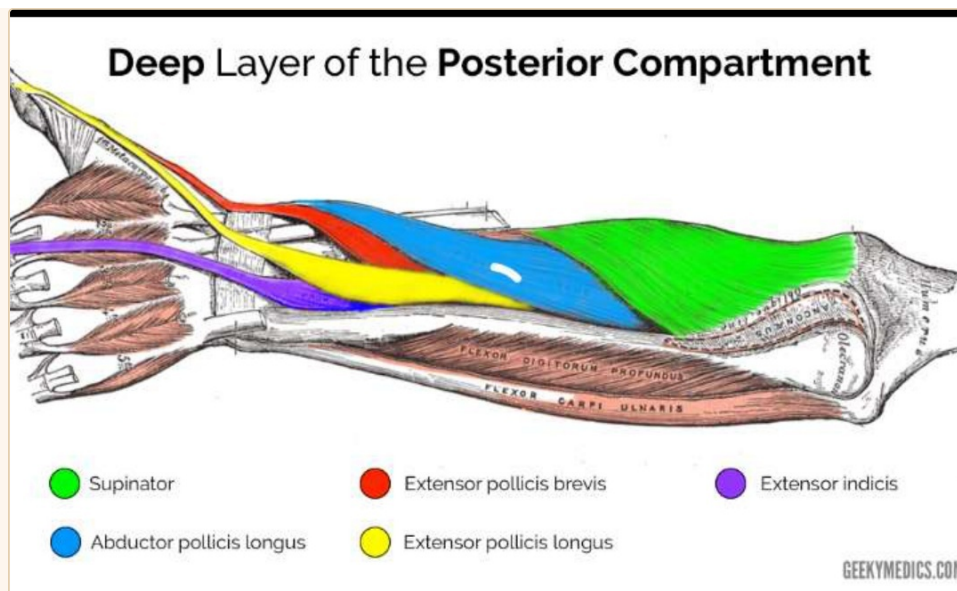
Alzheimer's Mode: Posterior forearm = radial nerve territory, no exceptions, no split-innervation traps — the "easy half" compared to the flexor side.

Superficial layer (common extensor origin: lateral epicondyle)



Muscle	Insertion	Action	Nerve
Brachioradialis (lateral-most)	Radial styloid process	Flexes the elbow in mid-pronation (the "imposter" of this compartment)	Radial
Extensor carpi radialis longus	Base of index metacarpal	Extension + abduction of wrist	Radial
Extensor carpi radialis brevis	Base of middle metacarpal	Extension of wrist	Posterior interosseous nerve
Extensor digitorum	Middle phalanges of fingers 2-5	Extension of hand + fingers	Posterior interosseous nerve
Extensor digiti minimi	Distal phalanx of little finger	Extension of little finger	Posterior interosseous nerve
Extensor carpi ulnaris (most medial)	Base of 5th metacarpal	Extension + adduction of wrist	Posterior interosseous nerve
Anconeus	Olecranon	Assists elbow extension	Radial nerve

Deep layer



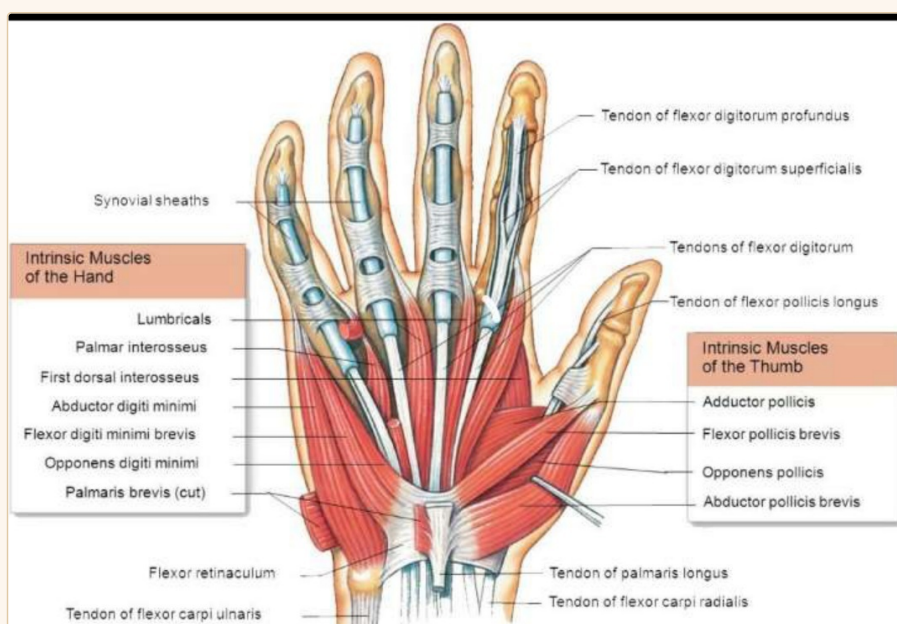
Muscle	Insertion	Action	Nerve
Supinator	Wraps around proximal radius	Supination	Posterior interosseous nerve
Abductor pollicis longus	Base of thumb metacarpal	Abduction/extension of thumb	Posterior interosseous nerve
Extensor pollicis brevis	Proximal phalanx of thumb	Extension of thumb	Posterior interosseous nerve
Extensor pollicis longus	Distal phalanx of thumb	Extension of thumb	Posterior interosseous nerve
Extensor indicis	Proximal phalanx of index	Independent extension of index finger	Posterior interosseous nerve

EXAM HACK

Brachioradialis is the imposter — lives in the extensor/radial nerve compartment but its ACTION is elbow flexion, strongest in mid-pronation (like lifting a mug). Every exam tests this contradiction. **Anatomical snuffbox borders:** lateral = Abductor pollicis longus + Extensor pollicis brevis tendons; medial = Extensor pollicis longus tendon; floor = scaphoid (snuffbox tenderness → suspect scaphoid fracture); the radial artery runs through it. **Lateral epicondylitis ("tennis elbow")** = inflammation at the common extensor origin, most often extensor carpi radialis brevis specifically.

Muscles of the Hand — Intrinsic Muscles

Alzheimer's Mode: 4 groups — *Thenar* (thumb pad), *Hypothenar* (little finger pad), *Central* (lumbricals + interossei), and *Adductor pollicis* (a group of one). Nerve rule: ulnar nerve owns almost everything; median nerve gets only a small VIP section.



Thenar muscles (all median nerve, recurrent branch) — mnemonic "AFO"

Muscle	Action
Abductor pollicis brevis	Abducts the thumb
Flexor pollicis brevis	Flexes the thumb
Opponens pollicis	Opposition (medial rotation) of the thumb

Hypothenar muscles (all ulnar nerve, deep branch) — same naming pattern, mirrored nerve

Muscle	Action
Abductor digiti minimi	Abducts the little finger
Flexor digiti minimi brevis	Flexes the little finger
Opponens digiti minimi	Opposition (lateral rotation) of the little finger

Single/central intrinsic muscles

Muscle	Action	Nerve
Adductor pollicis	Adducts the thumb	Ulnar (deep branch) — the odd one out among thumb intrinsics
Palmar interossei (4)	ADduct fingers toward the middle finger axis	Ulnar (deep branch)
Dorsal interossei (4)	ABduct fingers away from the middle finger axis	Ulnar (deep branch)
Lumbricals (4)	Flex MCP joints while extending IP joints simultaneously	Split: lateral 2 (index/middle) = Median; medial 2 (ring/little) = Ulnar

EXAM HACK

Thenar/hypothenar mirror trick: muscle NAMES mirror exactly (Abductor-Flexor-Opponens on both sides) but the NERVE flips — thumb side = median, little finger side = ulnar. **"PAD and DAB"** — Palmar interossei ADduct, Dorsal interossei ABduct. Lumbricals are the ONLY intrinsic hand muscles with split median/ulnar innervation — mirroring the same split-innervation pattern you already know from brachialis (arm) and flexor digitorum profundus (forearm), a recurring exam theme across the whole limb. Adductor pollicis is ulnar-supplied despite acting on the thumb — the classic "odd one out" among thumb muscles.

CLINICAL

Froment's sign (ulnar nerve palsy test): gripping paper between thumb and index finger — with weak adductor pollicis (ulnar), the patient compensates using flexor pollicis longus (median), flexing the thumb IP joint. **Ulnar claw hand:** hyperextension at MCP + flexion at IP of ring/little finger (loss of lumbricals 3&4). **Median claw/"ape hand"** (carpal tunnel syndrome, severe): thenar wasting, thumb can't oppose.

6. VASCULATURE OF THE UPPER LIMB

The blood supply begins as the subclavian artery (direct branch of the aortic arch on the left; from the brachiocephalic trunk on the right), continuing as the axillary artery, then the brachial artery, bifurcating into the ulnar and radial arteries.

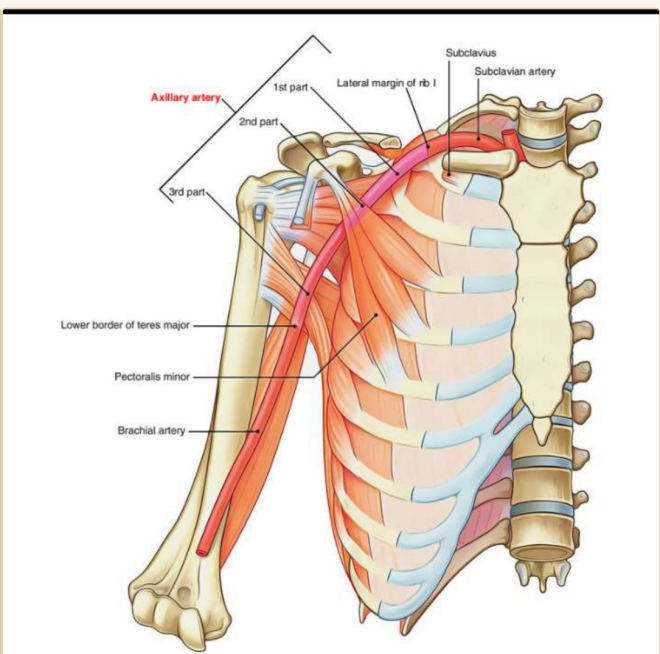
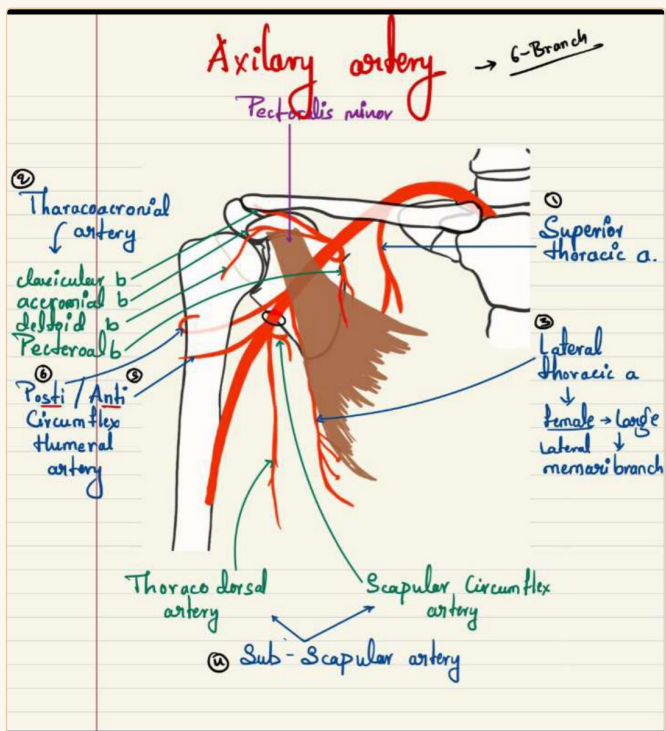
Subclavian Artery and Branches

Branch	Supplies
Vertebral artery (1st branch)	Ascends through cervical transverse foramina → brain
Thyrocervical trunk (2nd branch) — 4 sub-branches	Suprascapular a., Transverse cervical a. (superficial branch + deep branch = dorsal scapular a.), Inferior thyroid a., Ascending cervical a.
Internal thoracic artery	Descends to supply the chest wall
Costocervical trunk (last branch)	Deep cervical a. + Superior intercostal a. (1st two posterior intercostal arteries)

Axillary Artery and Branches

Alzheimer's Mode: Described in 3 parts relative to pectoralis minor, and the branch count matches the part number. "1-2-3."

Part	Branch(es)
1st part (above pec. minor) — 1 branch	Superior thoracic artery
2nd part (behind pec. minor) — 2 branches	Thoraco-acromial trunk (4 sub-branches: pectoral, acromial, clavicular, deltoid) + Lateral thoracic artery
3rd part (below pec. minor) — 3 branches	Subscapular a. (→ circumflex scapular + thoracodorsal aa.), Anterior circumflex humeral artery, Posterior circumflex humeral artery (larger diameter than anterior)



The axillary artery and its six branches, arranged in three parts around pectoralis minor.

EXAM HACK

6 branches total, mnemonic: **"Screw The Lawyer, Save A Patient"** — Superior Thoracic, Thoracoacromial, Lateral thoracic, Subscapular, Anterior circumflex humeral, Posterior circumflex humeral. The posterior circumflex humeral artery travels WITH the axillary nerve through the quadrangular space — paired structures, paired injury risk.

CLINICAL — COLLATERAL CIRCULATION

The subscapular artery's circumflex scapular branch joins an anastomotic network of vessels around the scapula (also fed proximally from the subclavian artery). If the subclavian or axillary artery is blocked anywhere between the first part of the subclavian and the third part of the

axillary artery, this scapular anastomosis can serve as a collateral pathway, keeping the upper limb adequately perfused.

Brachial Artery and Branches

The axillary artery becomes the brachial artery at the lower margin of teres major; it terminates at the neck of the radius by bifurcating into the ulnar and radial arteries. It lies medial to the biceps tendon in the cubital fossa, where its pulse is easily palpated.

- **Profunda brachii artery:** runs posteriorly through the triangular interval, alongside the radial nerve in the radial/spiral groove; gives the radial collateral artery.
- **Superior ulnar collateral artery:** descends posterior to the medial epicondyle → anastomoses with the posterior recurrent ulnar artery.
- **Inferior ulnar collateral artery:** anastomoses posteriorly with the radial collateral artery and anteriorly with the anterior recurrent ulnar artery.

EXAM HACK

The elbow anastomotic network links: profunda brachii's radial collateral, superior + inferior ulnar collateral arteries, and the anterior + posterior recurrent ulnar arteries. This collateral network matters clinically if the brachial artery is occluded or ligated near the elbow — collateral flow can maintain forearm perfusion.

Ulnar Artery and Branches

- **Anterior ulnar recurrent artery:** ascends to anastomose with the inferior ulnar collateral artery.
- **Posterior ulnar recurrent artery:** ascends posterior to the medial epicondyle → anastomoses with the superior ulnar collateral artery.
- **Common interosseous artery:** splits into anterior interosseous artery (anterior forearm) + posterior interosseous artery (posterior forearm).
- **Volar carpal + dorsal carpal branches** at the wrist.

Radial Artery and Branches

- **Radial recurrent artery:** ascends to anastomose with profunda brachii.
- **Palmar carpal artery:** anastomoses with the ulnar artery's volar carpal branch.

Arteries of the Hand

Alzheimer's Mode: Two palmar arches with opposite dominance, plus a dorsal arch — "Superficial = Ulnar-dominant, Deep = Radial-dominant."

Arch	Formed by	Supplies
Superficial palmar arch	Ulnar artery (main) + superficial branch of radial artery (completes it)	Digital arteries → fingers
Deep palmar arch	Radial artery (main) + deep branch of ulnar artery (completes it)	Deep palm muscles; gives metacarpal branches anastomosing with the superficial arch's digitals
Princeps pollicis artery	Deep palmar branch of radial artery	Thumb (both sides)
Radialis indicis artery	Deep radial palmar artery (before forming deep arch)	Lateral side of index finger
Dorsal carpal arch	Dorsal volar artery + dorsal radial carpal artery	Digital branches on the dorsum of the hand

EXAM HACK

Don't confuse the two palmar arches' dominance: **Superficial = ULNAR-dominant; Deep = RADIAL-dominant**. The **dorsal carpal arch** is a separate third arch on the back of the wrist — don't lump it with the two palmar arches. **Allen's test** checks patency of both radial and ulnar arteries plus the completeness of the palmar arch anastomosis: occlude both, release one at a time, watch colour return.

Venous Drainage of the Upper Limb

Alzheimer's Mode: Two systems — deep veins shadow the arteries with the same names; superficial veins run their own course just under the skin, and it's the superficial pair (cephalic + basilic) that examiners and phlebotomists both care about.

- **Deep veins** — venae comitantes accompanying the named arteries (radial, ulnar, brachial, axillary, subclavian veins). The

axillary vein forms from the union of the brachial and basilic veins at the inferior border of teres major, and (like the artery) is described in three parts; its first part becomes the subclavian vein at the lateral border of the first rib.

- **Cephalic vein** (lateral side): runs up the lateral arm to the deltopectoral groove and drains into the axillary vein.
- **Basilic vein** (medial side): pierces the deep fascia in the arm to join the venae comitantes and form the axillary vein.
- **Median cubital vein**: connects the cephalic and basilic veins over the cubital fossa — the vein most commonly used for venepuncture and IV cannulation.

CLINICAL

Axillary vein thrombosis can occasionally occur when an accessory muscular band (an "axillary arch") compresses the vein, producing sudden arm pain, swelling or skin discolouration. Deep veins carry more valves than superficial veins, helping drive blood back toward the heart.

7. LYMPHATIC DRAINAGE OF THE UPPER LIMB

Alzheimer's Mode: Lymph follows veins home — superficial lymphatics track the superficial veins, deep lymphatics track the deep vessels, and almost everything eventually funnels into one basin: the axilla.

- Superficial lymphatics generally follow the superficial veins (cephalic, basilic); deep lymphatics follow the deep vessels.
- Most lymph from the hand and forearm drains toward a small group of **cubital (supratrochlear) nodes** near the medial epicondyle, then continues up the arm alongside the basilic vein.
- Nearly all upper limb lymph ultimately drains into the **axillary lymph nodes** (about 20–30 nodes in five groups: anterior/pectoral, posterior/subscapular, lateral/humeral, central, apical), which in turn drain into the subclavian lymph trunk.

EXAM HACK

Five axillary node groups, easy to remember by position: **anterior (pectoral)** along the lower border of pec major, **posterior (subscapular)** along the subscapular vessels, **lateral (humeral)** along the axillary vein, **central** embedded in axillary fat receiving the other three, **apical** at the apex — the final common pathway into the subclavian trunk. "Central collects, apical exits."

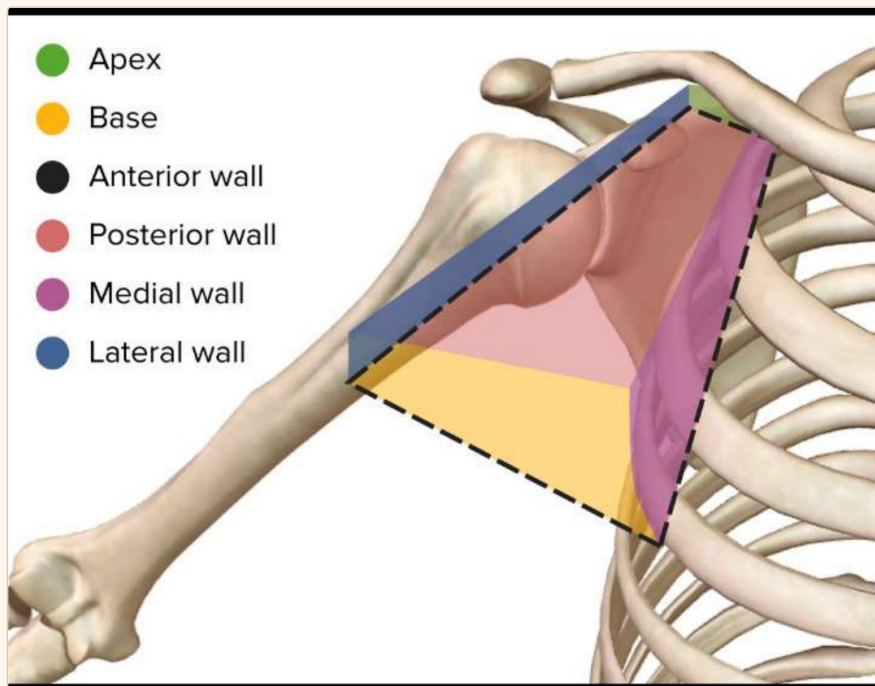
CLINICAL

Because the breast and the whole upper limb share the same axillary drainage basin, infection in the hand/arm and breast pathology can BOTH present as palpable axillary lymphadenopathy — this shared basin is exactly why axillary node staging matters so much in breast cancer (see Section 8).

8. THE AXILLA

Alzheimer's Mode: A four-sided pyramid tucked between the arm and the chest wall — picture a tent with its apex pointing up into the root of the neck and its base forming the visible armpit. Almost every major nerve and vessel of the upper limb passes through this one passageway.

Part	Formed by
Apex (cervico-axillary canal)	Clavicle (anteriorly), superior border of scapula (posteriorly), outer border of 1st rib (medially)
Base (floor)	Skin, superficial fascia and the axillary fascia — the visible armpit
Anterior wall	Pectoralis major and minor, subclavius, clavipectoral fascia
Posterior wall	Subscapularis, teres major, latissimus dorsi
Medial wall	Serratus anterior, overlying ribs 1-4 and intercostal muscles
Lateral wall	Intertubercular (bicipital) groove of the humerus



The axilla — apex, base and four walls.

Contents of the Axilla

- Axillary artery and its 6 branches (three parts, related to pectoralis minor — see Section 6).
- Axillary vein and its tributaries — lies anteromedial to the artery.
- Cords and terminal branches of the brachial plexus, surrounding the axillary artery.
- Axillary lymph nodes — the main drainage site for the breast and upper limb.
- Fat and loose connective (areolar) tissue filling the remaining space.

EXAM HACK — THE NAVEL RULE

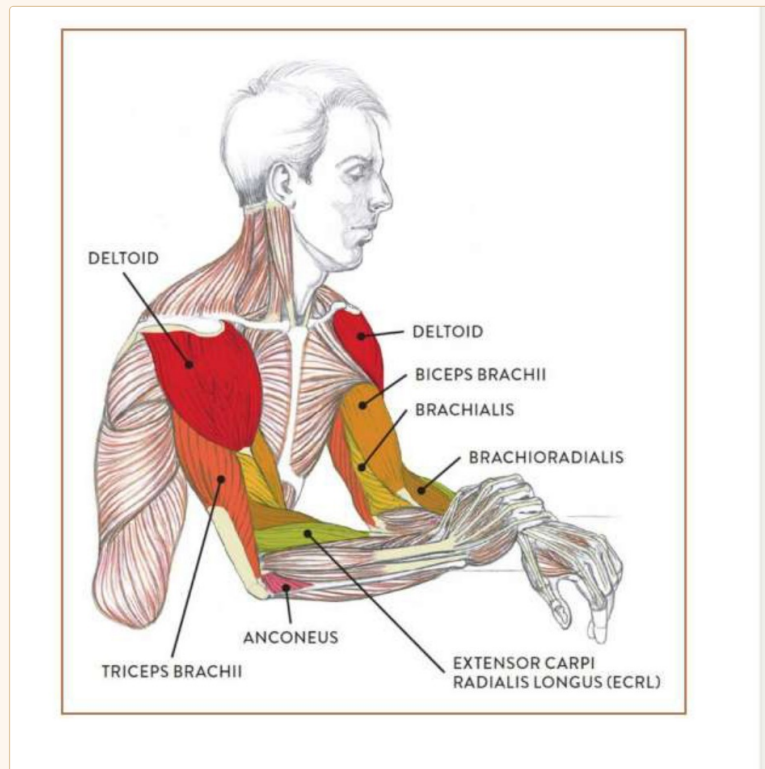
Most neurovascular compartments (axilla, cubital fossa, femoral triangle) can be remembered by the same content pattern: **NAVEL** — **N**erves, **A**rteries, **V**eins, **E**mpy space/fat, **L**ymphatics. Once you know this order applies to the axilla, the same pattern carries straight into the cubital fossa preview in Section 10 — one mnemonic, several spaces.

CLINICAL

Because pectoralis minor crosses the anterior wall and physically divides the axillary artery into its three parts, it is also the landmark used to define the **3 levels of axillary lymph nodes** in breast cancer staging (level I below/lateral to pec minor, level II behind it, level III above/medial to it) — the single most repeated axilla fact in clinical anatomy. Axillary node clearance during breast surgery risks the **long thoracic nerve** (winged scapula) and the **thoracodorsal nerve** (weak latissimus dorsi pull), both of which run close to the posterior/medial walls.

9. SURFACE ANATOMY OF THE UPPER LIMB

The palpable bony points, muscle bellies and pulses you can find on a real person underpin clinical examination and are a favourite source of quick OSCE-style questions.



- **Palpable bony points:** the whole shaft of the clavicle, the acromion and spine of scapula, the medial and lateral epicondyles of the humerus (with the olecranon between them forming a triangle when the elbow is flexed), the head of ulna and styloid processes of radius/ulna at the wrist.
- **Pulses:** the **brachial pulse** (medial to the biceps tendon in the cubital fossa), the **radial pulse** (lateral wrist, between the brachioradialis tendon and the radius – the most commonly checked pulse), and the **ulnar pulse** (medial wrist, less reliably felt).
- **Veins for cannulation:** the cephalic vein (lateral forearm/arm) and basilic vein (medial forearm/arm), joined by the median cubital vein at the elbow – the standard site for venepuncture.
- The **anatomical snuffbox** (seen on thumb extension) and the "bracelet" skin creases at the wrist are also useful surface landmarks (see Section 10).

EXAM HACK

Olecranon + medial epicondyle + lateral epicondyle form a triangle in flexion but sit in a straight line in full extension – a quick bedside check for a supracondylar or elbow dislocation injury (the triangle is disrupted only in true joint dislocation, not in a supracondylar fracture, where the triangle relationship is preserved).

10. TUTORIAL PREVIEW — CUBITAL FOSSA, CARPAL TUNNEL & SNUFFBOX

These three spaces are usually taught in tutorial in full depth — treat the notes below as a tight orientation you can expand on later, built on the same NAVEL logic as the axilla.

Cubital Fossa

A triangular depression on the anterior aspect of the elbow, bounded by brachioradialis (lateral), pronator teres (medial), and an imaginary line between the epicondyles (base). From lateral to medial it contains the biceps tendon, brachial artery, and median nerve.

EXAM HACK

Mnemonic (lateral → medial): "**TAN**" — Tendon (biceps), Artery (brachial), Nerve (median). The radial nerve lies just OUTSIDE the lateral border, not inside the fossa itself — a classic trap that adds a fourth structure where only three belong.

Carpal Tunnel

An osteofibrous tunnel on the palmar wrist — floor and sides formed by the carpal bones (carpal arch), roof formed by the flexor retinaculum. It transmits the median nerve and the tendons of flexor digitorum superficialis, flexor digitorum profundus, and flexor pollicis longus.

CLINICAL

Compression here causes **carpal tunnel syndrome** — numbness/tingling in the lateral 3½ fingers and thenar weakness, sparing the palm (the palmar cutaneous branch of the median nerve leaves before the tunnel and runs superficial to the retinaculum).

Anatomical Snuffbox

A triangular depression on the radial side of the dorsum of the wrist, seen when the thumb is extended; bounded by the tendons of abductor pollicis longus/extensor pollicis brevis (anterior/lateral border) and extensor pollicis longus (posterior/medial border). The floor is the scaphoid and trapezium, and the radial artery crosses through it.

CLINICAL

Snuffbox tenderness after a fall is important because scaphoid fractures classically present with tenderness exactly here — and, given the scaphoid's retrograde blood supply (Section 2), a missed fracture risks avascular necrosis.

11. MASTER HACK SHEET — LAST-MINUTE REVIEW

Every mnemonic, trap, and high-yield rule from this compendium, compressed onto one section. Use this the night before the exam.

OSTEOLOGY

Clavicle: dual ossification (intramembranous shaft + endochondral medial epiphysis, last to fuse ~25yo); only bone from 2 primary centres; most fractured bone; breaks at middle-lateral third junction.

Scapula: "3-3-2-3" (borders-angles-surfaces-processes); suprascapular NOTCH = nerve only, GREATER notch = artery+nerve.

Humerus 3-nerve rule: surgical neck→axillary; radial groove→radial; medial epicondyle→ulnar.

Ulna=fixed pivot, Radius=rotates. MUGR: Monteggia=Ulna(proximal); Galeazzi=Radius(distal).

Carpals: "Some Lovers Try Positions That They Cannot Handle." Scaphoid=most fractured, retrograde supply→AVN risk. Lunate=most dislocated.

JOINTS & LIGAMENTS

Coracoclavicular ligament (conoid+trapezoid) = TRUE AC joint stabilizer.

Glenohumeral: IGL=primary anterior-dislocation restraint (hammock); coracoclavicular ligament ≠ true joint ligament (no humeral attachment); rotator cuff=dynamic, labrum/capsule/GHLs=static.

Subacromial bursa = most clinically important, shares space with supraspinatus (impingement).

Elbow & IP joints = pure hinge. Glenohumeral = ONLY ball-and-socket in upper limb. 1st CMC = saddle joint.

SCAPULAR/SHOULDER MUSCLES

Trapezius = CN XI (only cranial-nerve-supplied muscle here).

Levator scapulae + both rhomboids = dorsal scapular nerve.

Serratus anterior + trapezius = force couple for overhead abduction.

SITS = rotator cuff; 3 of 4 → greater tubercle + suprascapular n. (except teres minor=axillary n.); subscapularis=odd one out (lesser tubercle, medial rotation).

Abduction relay: 0-15°=supraspinatus, 15-90°=deltoid, >90°=trapezius+serratus anterior.

Pec major moves the ARM; pec minor + subclavius move the SCAPULA/CLAVICLE — don't lump together.

Intertubercular groove trio: pec major (lateral lip), lat dorsi (floor), teres major (medial lip).

PLANES, AXES & TERMS

Perpendicular partners: Sagittal plane→transverse axis. Coronal plane→sagittal axis. Transverse plane→longitudinal axis.

Only the MEDIAN plane gives equal halves (parasagittal = unequal).

Circumduction = combo of flexion+extension+abduction+adduction, NOT a separate movement.

Elevation/depression also occurs at the mandible, not just the shoulder.

Anatomical position = palms face ANTERIORLY (fully supinated).

Proximal/distal ≠ medial/lateral — proximal is about trunk distance, not midline.

BRACHIAL PLEXUS

"Randy Travis Drinks Cold Beer" = Roots→Trunks→Divisions→Cords→Branches.

Trunks: Sup=C5,6; Mid=C7; Inf=C8,T1.

Posterior cord=only cord from every root; gives Axillary+Radial.

Lateral cord alone→Musculocutaneous; Medial cord alone→Ulnar; BOTH→Median.

Pre-trunk branches: Long thoracic (serratus ant.), Dorsal scapular (rhomboids/lev.scap), N. to subclavius.

Erb's=upper trunk=waiter's tip. Klumpke's=lower trunk=claw hand ±Horner's.

ARM & FOREARM MUSCLES

Brachialis = true prime elbow flexor (all positions); biceps = better supinator.

Brachialis = dual nerve (musculocutaneous + radial) — "odd one out" #1.

Anterior forearm = median-dominant; ulnar nerve gets exactly 1½ muscles (FCU + medial ½ FDP).

FDP = dual nerve (ulnar + AIN) — "odd one out" #2, mirrors brachialis.

Posterior forearm = 100% radial nerve, no exceptions.

Brachioradialis = imposter (extensor compartment, but FLEXES elbow).

Snuffbox: APL+EPB lateral border, EPL medial border, scaphoid floor, radial artery runs through it.

HAND INTRINSICS

Thenar (AFO) = median n. Hypothenar (mirror names) = ulnar n.

Adductor pollicis = ulnar (odd one out among thumb muscles).

Lumbricals = split: lateral 2=median, medial 2=ulnar — 3rd "odd one out" split-innervation pattern (after brachialis, FDP).

PAD = Palmar ADduct; DAB = Dorsal ABduct.

Froment's sign = ulnar palsy compensation via median-supplied FPL.

VASCULATURE & LYMPHATICS

Axillary artery "1-2-3" branches per part (split by pec minor). Mnemonic: "Screw The Lawyer, Save A Patient."

Posterior circumflex humeral a. travels with axillary nerve (quadrangular space).

Elbow anastomoses: profunda brachii's radial collateral + sup/inf ulnar collateral + ant/post recurrent ulnar arteries.

Palmar arches: Superficial=ULNAR-dominant; Deep=RADIAL-dominant. Dorsal carpal arch = separate third arch.

Cephalic (lateral) + basilic (medial) veins joined by median cubital v. at the elbow — standard venepuncture site.

Axillary nodes: pectoral/subscapular/humeral→central→apical. Levels I-III defined by pec minor — shared basin with the breast.

Allen's test = checks radial+ulnar patency + arch anastomosis.

AXILLA & SPECIAL SPACES

Axilla = apex/base + 4 walls (ant=pec major/minor, post=subscapularis/teres major/lat dorsi, med=serratus anterior, lat=intertubercular groove).

NAVEL rule (Nerves-Arteries-Veins-Empty space-Lymphatics) applies to axilla AND cubital fossa.

Cubital fossa "TAN" lateral→medial: Tendon, Artery, Nerve (radial n. lies just outside).

Carpal tunnel: median n. + FDS/FDP/FPL tendons under flexor retinaculum — palmar cutaneous branch spared (leaves before the tunnel).

RECURRING "ODD ONE OUT" THEME

Split/dual innervation appears 3 times across the limb — spot the pattern:

- 1) Brachialis (arm) — musculocutaneous + radial
- 2) Flexor digitorum profundus (forearm) — ulnar + anterior interosseous
- 3) Lumbricals (hand) — median (lateral 2) + ulnar (medial 2)

Recognising this repeating structure is worth more than memorizing each fact in isolation.

CLINICAL QUICK-FIRE

Surgical neck fx → axillary n. → deltoid palsy, regimental badge numbness.

Mid-shaft humeral fx → radial n. → wrist drop.

Supracondylar fx (children) → brachial artery + median n.

Anterior shoulder dislocation → Bankart (IGL/labrum) + Hill-Sachs (humeral head) lesions.

Colles' (dorsal) vs Smith's (volar) distal radius fractures.

Scaphoid fx → snuffbox tenderness → AVN risk (retrograde supply).

Long thoracic n. → winged scapula. Accessory n. → trapezius palsy, shoulder droop.

Axillary node clearance → long thoracic + thoracodorsal nerves at risk (shared breast/limb lymphatic basin).